Christof Teuscher

ECE 410/510: Hardware for AI and ML, Spring 2026

Emerging devices and technologies: What's next?

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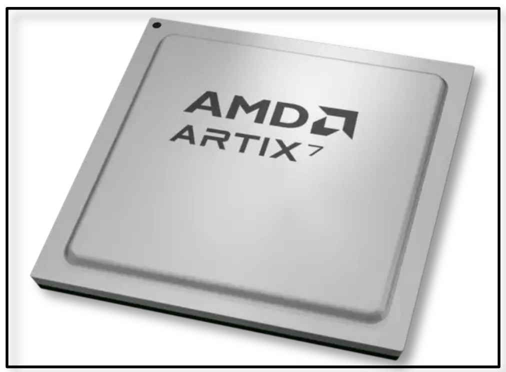

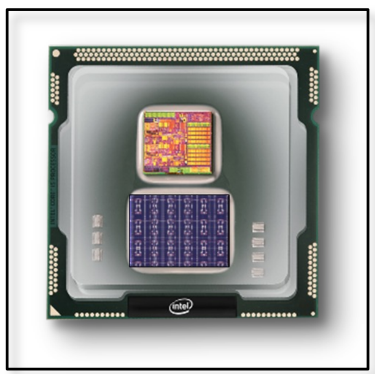

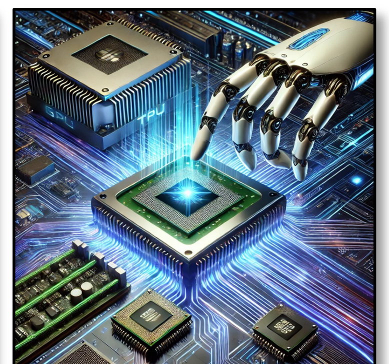

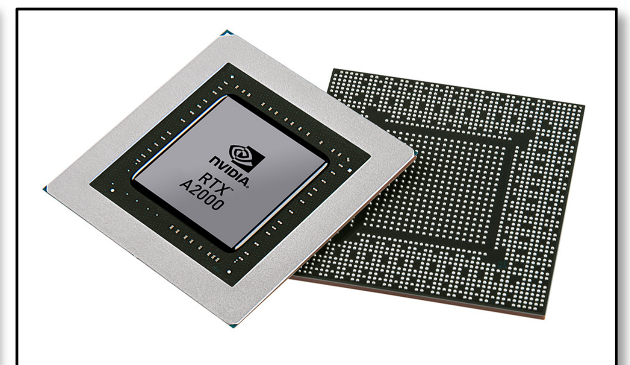

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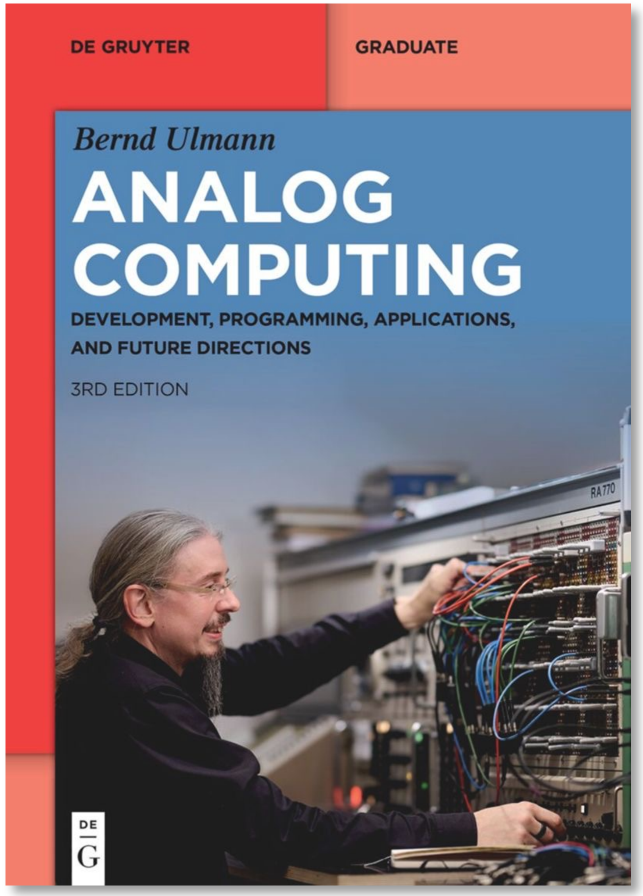
What other options do we have to accelerate AI/ML algorithms?

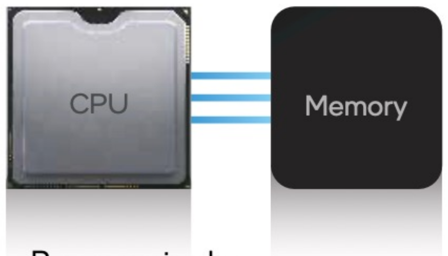
More of the same: physics!

- Compute closer to the physics.
- Compute with the physics.
- Analog computing is back!



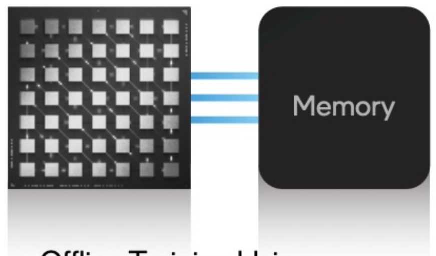
Unconventional computing

Conventional Computing



- Programming by Encoding Algorithms
- Synchronous Clocking
- Sequential Threads of Control

Parallel Computing



- Offline Training Using Labeled Datasets
- Synchronous Clocking
- Parallel Dense Compute

Neuromorphic Computing

- Learn On-the-Fly Through Neuron Firing Rules
- Asynchronous Event-Based Spikes
- Parallel Sparse Compute

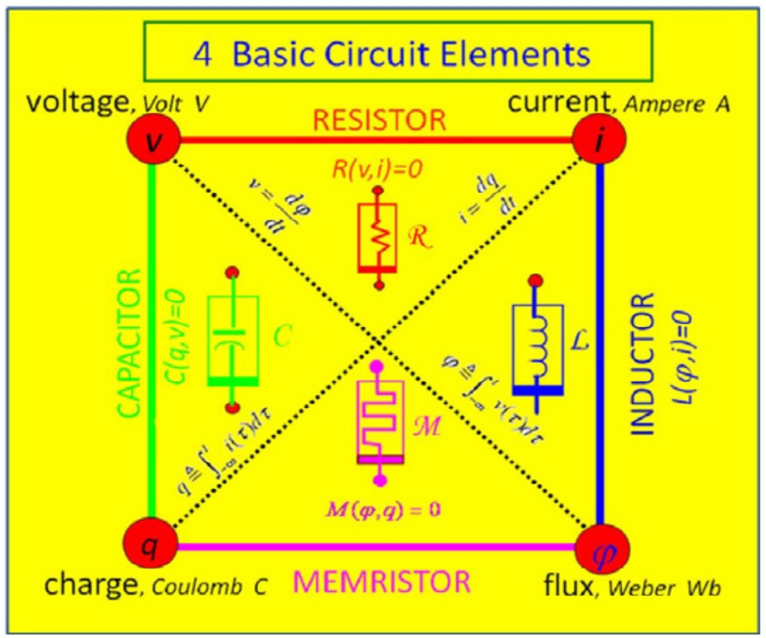
Unconventional Computing

?

Technology Brief | Taking Neuromorphic Computing to the Next Level with Loihi 2

Mem-elements

- 1971: Chua predicted missing circuit elements



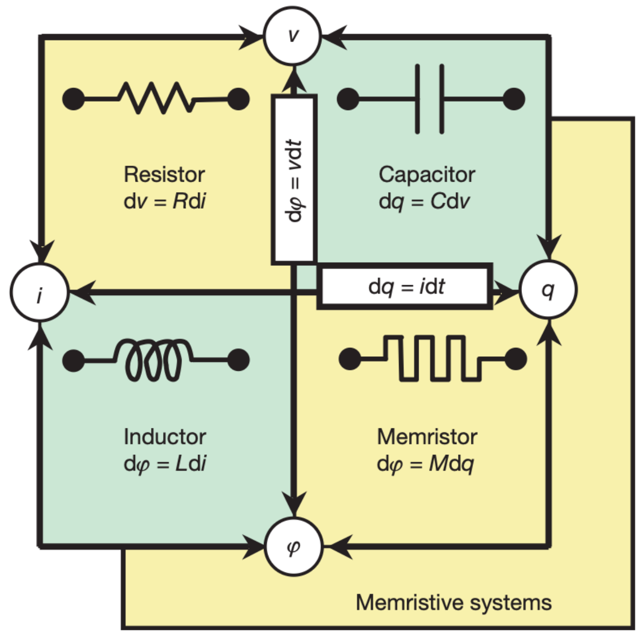


Figure 1 | The four fundamental two-terminal circuit elements: resistor, capacitor, inductor and memristor. Resistors and memristors are subsets of a more general class of dynamical devices, memristive systems. Note that R, C, L and M can be functions of the independent variable in their defining equations, yielding nonlinear elements. For example, a charge-controlled memristor is defined by a single-valued function $M(q)$.

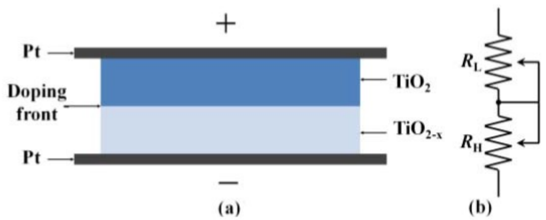
Memristor:

- Variable resistance that depends on the history of current that has flowed through the device.
- Ability to "remember" its previous resistance state even when power is turned off.
- Non-volatile memory properties (retains information without power)

HP: TiO_2 devices

https://doi.org/10.1038/nature06932

Memristors



TiO$_2$ memristor. (a) Structure. (b) Equivalent circuit.

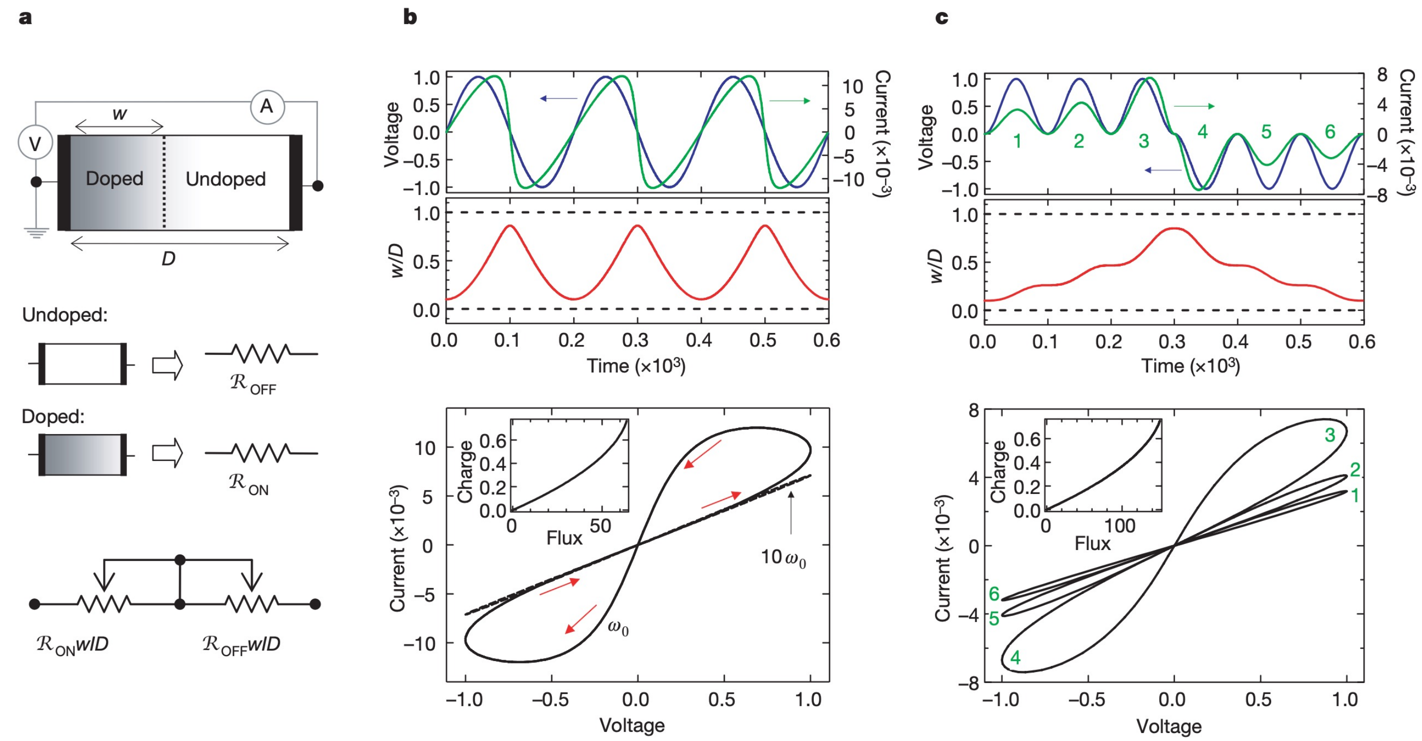


Figure 2 | The coupled variable-resistor model for a memristor. a, Diagram with a simplified equivalent circuit. V, voltmeter; A, ammeter. **b**, **c**, The applied voltage (blue) and resulting current (green) as a function of time t for a typical memristor. In **b** the applied voltage is $v_0 \sin(\omega_0 t)$ and the resistance ratio is $\mathcal{R}_{OFF}/\mathcal{R}_{ON} = 160$, and in **c** the applied voltage is $\pm v_0 \sin^2(\omega_0 t)$ and $\mathcal{R}_{OFF}/\mathcal{R}_{ON} = 380$, where v_0 is the magnitude of the applied voltage and ω_0 is the frequency. The numbers 1–6 label successive waves in the applied voltage and the corresponding loops in the i–v curves. In each plot the axes are dimensionless, with voltage, current, time, flux and charge expressed in units of $v_0 = 1$ V, $i_0 \equiv v_0/\mathcal{R}_{ON} = 10$ mA, $t_0 \equiv 2\pi/\omega_0 \equiv D^2/\mu_V v_0 = 10$ ms, $v_0 t_0$ and $i_0 t_0$, respectively. Here i_0 denotes the maximum possible current through the device, and t_0 is the shortest time required for linear drift of dopants across the full device length in a uniform field v_0/D, for example with $D = 10$ nm and $\mu_V = 10^{-10}$ cm^2 s^{-1} V^{-1}. We note that, for the parameters chosen, the applied bias never forces either of the two resistive regions to collapse; for example, w/D does not approach zero or one (shown with dashed lines in the middle plots in **b** and **c**). Also, the dashed i–v plot in **b** demonstrates the hysteresis collapse observed with a tenfold increase in sweep frequency. The insets in the i–v plots in **b** and **c** show that for these examples the charge is a single-valued function of the flux, as it must be in a memristor.

https://doi.org/10.1038/nature06932

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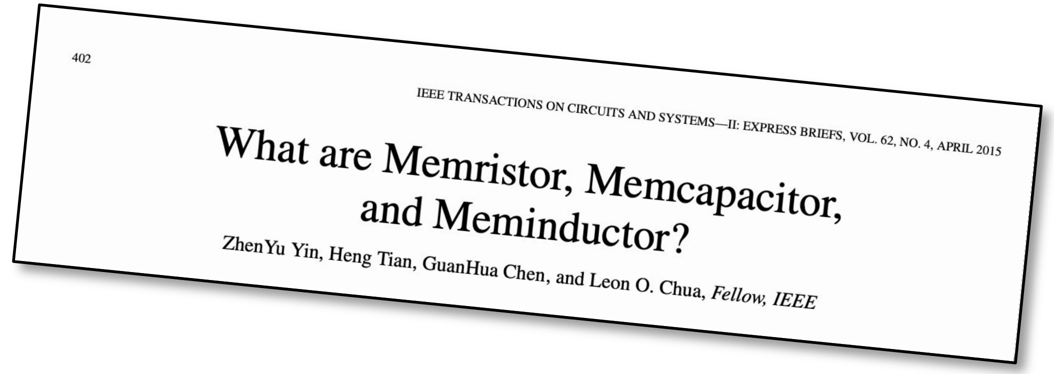
Memcapacitor and meminductor

Feature	Memristor	Memcapacitor	Meminductor
Basic Definition	Memory resistor	Memory capacitor	Memory inductor
Conventional Counterpart	Resistor	Capacitor	Inductor
Controls/Responds To	Current history	Voltage history	Flux history
State Variable	Resistance (memristance)	Capacitance (memcapacitance)	Inductance (meminductance)
Fundamental Relationship	Links charge (q) and magnetic flux (φ)	Links charge (q) and voltage (v)	Links current (i) and magnetic flux (φ)
Energy Storage	Minimal/none	Electric field	Magnetic field
Primary Physical Mechanisms	Ion migration, filament formation, phase changes	Polarization changes, ionic redistribution	Magnetic domain reorientation
State Change Method	Apply current	Apply voltage	Apply magnetic flux
First Theoretical Proposal	1971 (Leon Chua)	2009 (Di Ventra, Pershin, Chua)	2009 (Di Ventra, Pershin, Chua)
First Physical Implementation	2008 (HP Labs)	Limited experimental demonstrations	Limited experimental demonstrations
Development Status	Multiple working prototypes	Mostly theoretical, some prototypes	Mostly theoretical, fewer prototypes
Key Applications	Non-volatile memory, neuromorphic computing	Signal processing, tunable circuits	Power systems, signal filters
Characteristic Property	Non-linear resistance with memory	Non-linear capacitance with memory	Non-linear inductance with memory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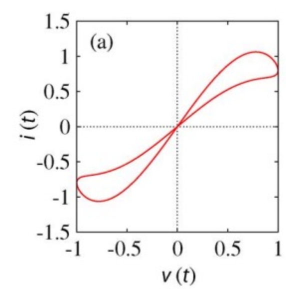
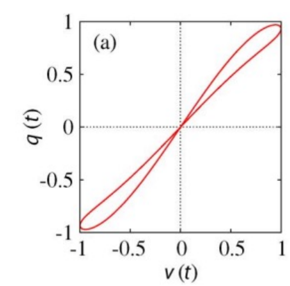
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS, VOL. 62, NO. 4, APRIL 2015

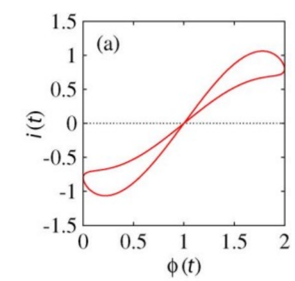
What are Memristor, Memcapacitor, and Meminductor?

ZhenYu Yin, Heng Tian, GuanHua Chen, and Leon O. Chua, *Fellow, IEEE*

- Memristor state change: apply an electric current through the device.
- Memcapacitors store energy in electric fields and respond to voltage history rather than current history.
- Meminductor state change: create magnetic flux.







DOI: 10.1109/TCSII.2014.2387653

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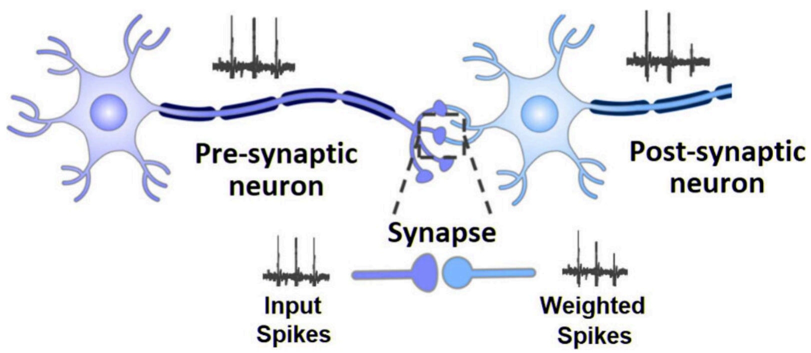
Memristors can implement STDP

- STDP emerges naturally from device physics without complex circuits.
- STDP: The time difference between spikes determines magnitude of change (closer timing = stronger effect)
- STDP refines Hebbian learning by adding precise timing relationships. Unsupervised learning!
- Memristors typically have threshold voltages required for changing state.
- **Only when spikes overlap sufficiently does voltage exceed the threshold.**



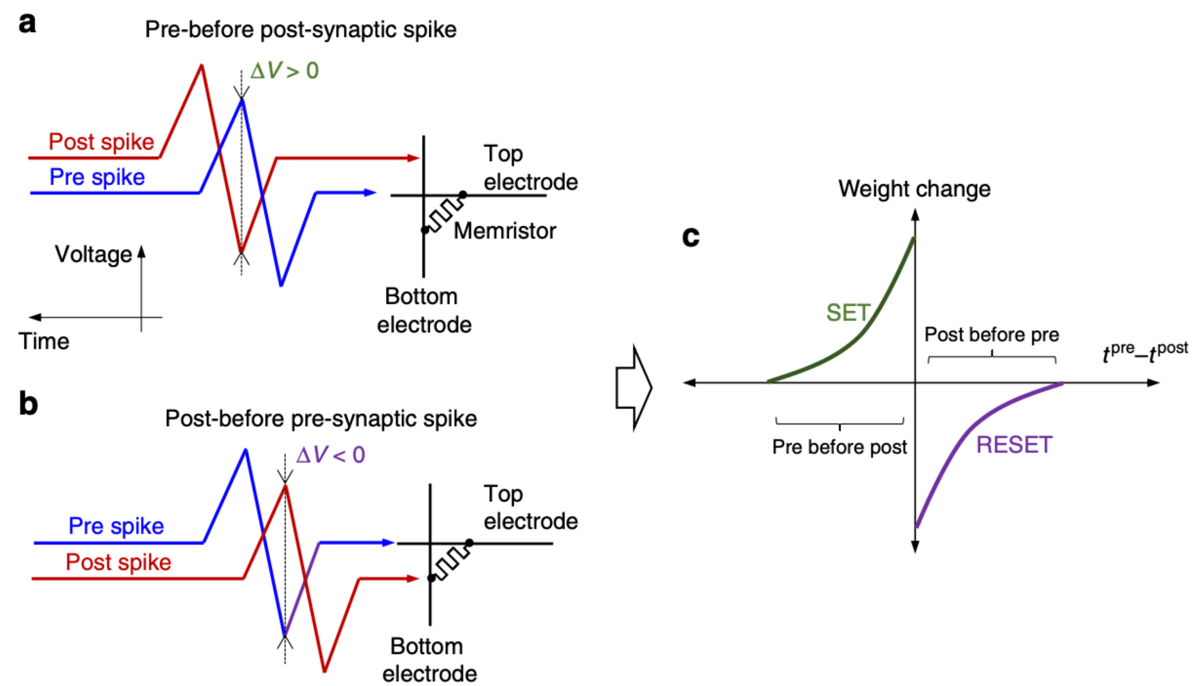


Fig. 1 STDP implementation. Relative timing of the pre- and post-synaptic spikes for performing **a** potentiation and **b** depression of the synaptic weight for the memristor-based implementation[11], resulting in **c** the specific STDP, which is commonly observed in biology

https://doi.org/10.1038/s41467-018-07757-y

What is one problem of memristors?

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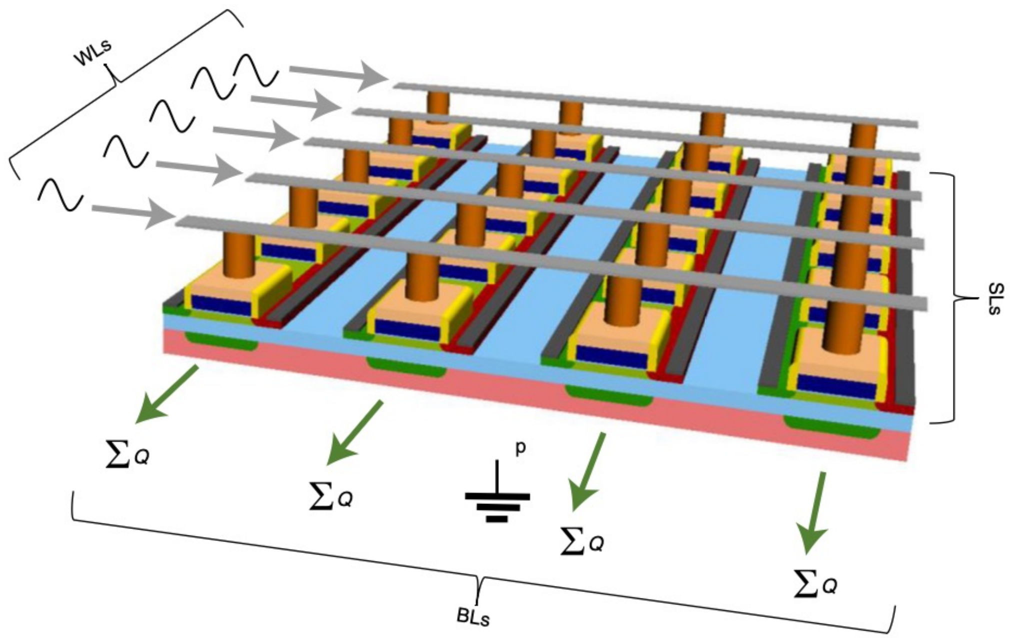
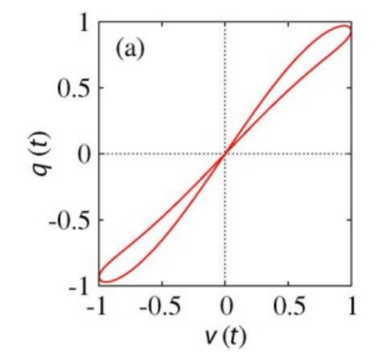
Memcapacitors

- Memcapacitors offer several unique benefits, but one of their most significant advantages is their exceptional energy efficiency in dynamic computing applications.
- Memcapacitors can implement STDP too (through charge).





https://doi.org/10.1038/s41928-021-00649-y

Issues with mem-devices

- Device-to-device variation
- Cycle-to-cycle variation
- Limited write cycles (10^6-10^9 compared to CMOS $>10^{15}$)
- Retention issues / state drift
- Programming complexity (requires precise pulses)
- Limited switching speeds
- Lack of design tools

https://doi.org/10.1038/s41928-021-00649-y

Reconfigurable perovskite nickelate electronics for AI

- **Problem:** AI hardware is static. Networks suffer catastrophic forgetting as new data arrives.
- **Idea:** One device that becomes a resistor, capacitor, neuron, or synapse. On demand!
- **Material:** Perovskite $NdNiO_3$ (NNO). H^+ doping drives a room-temperature electronic phase transition.
- **Knob:** Single-shot voltage pulses redistribute protons in the lattice, nonvolatile, voltage-controlled.
- **Result:** Same fabricated device, reprogrammed post-fab into different neuromorphic primitives.

https://doi.org/10.1126/science.abj7943

One device, four functions, reconfigured by a voltage pulse

(i) Linear Resistor
Pristine / weakly doped nickelate (NNO)
Linear I–V under cyclic sweep

(ii) Capacitor
Intermediate H-doping
I–V loop → stored energy
Memcapacitive behavior

(iii) Spiking Neuron
Stochastic firing at threshold
–0.45 V/μm pulse, 1 ms
Reset at +0.45 V/μm

(iv) Analog Synapse
Continuous voltage sweeps
Gradual R increase
Analog weight updates

https://doi.org/10.1126/science.abj7943

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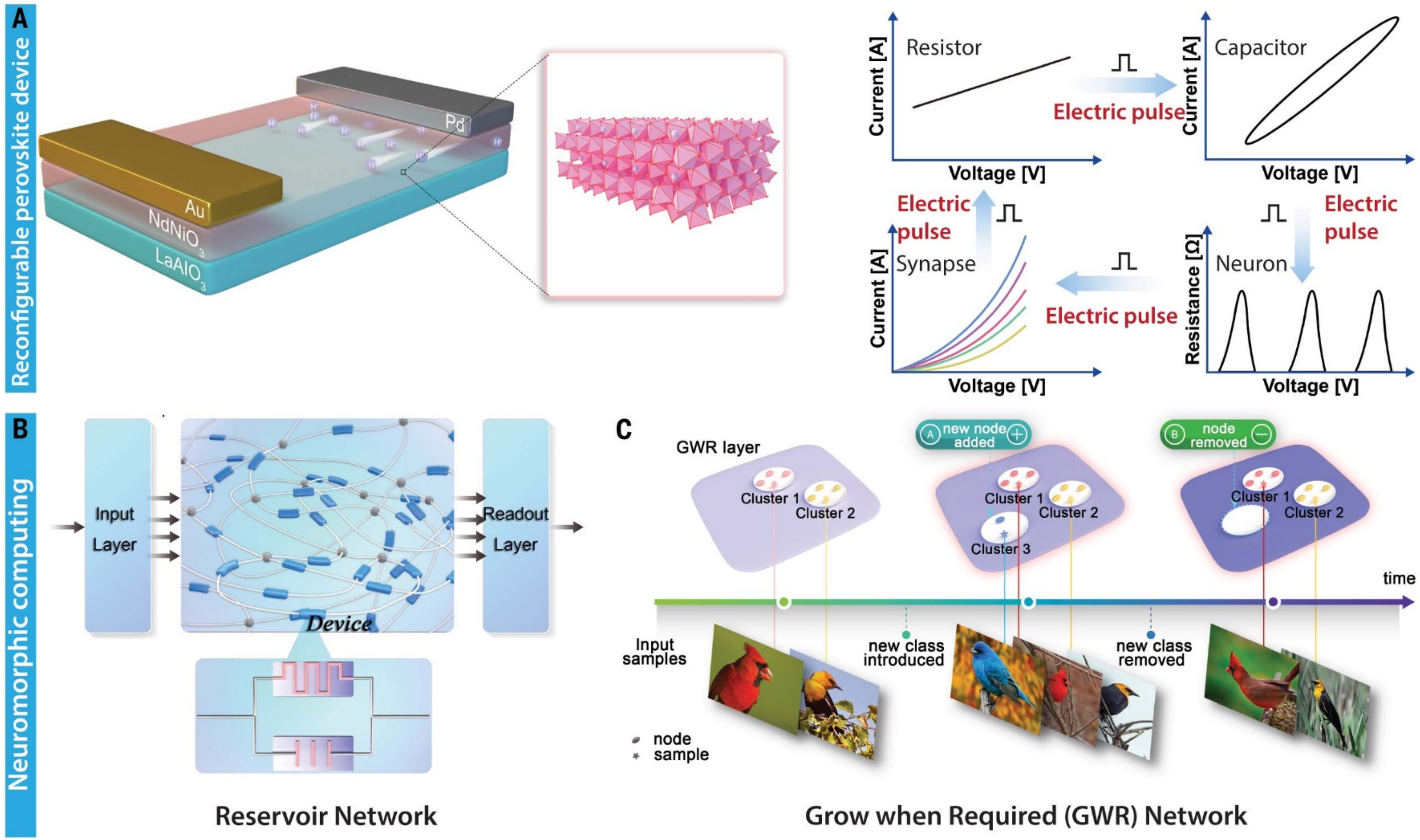


Fig. 1. Reconfigurable perovskite devices. (**A**) Schematic of hydrogen-doped perovskite nickelate as a versatile reconfigurable platform that can be electrically transformed between neurons, synapses, and memory capacitors to enable adaptive neuromorphic computing. By applying electric pulses, the hydrogen ions in the nickelate lattice can occupy metastable states and enable distinct functionalities. (**B**) Schematic of a generic RC framework. An input layer distributes the signals into the reservoir, which projects the inputs into a high-dimensional space. Here, the reservoir is built randomly from programmable devices with memory. No training happens in the reservoir; only the linear readout layer is trained by a simple gradient descent algorithm. The role of the readout layer is to map the high-dimensional dynamics of the reservoir to the output states. (**C**) Schematic of GWR networks. As the network is shown various classes of data, it maps high-dimensional data to a low-dimensional map field to perform clustering on the classes. When a new class is added to the input stream, the network can detect the new input and grow in size by adding network nodes to accommodate it. Additionally, if any of the classes do not appear in the input stream for a long time, the corresponding nodes become inactive, saving resources.

From device physics to AI: two demonstrations

Reservoir computing

- Built from H-NNO memory capacitors
- Digit recognition (MNIST)
- ECG heartbeat classification
- Excellent performance from raw device dynamics

Dynamic GWR network

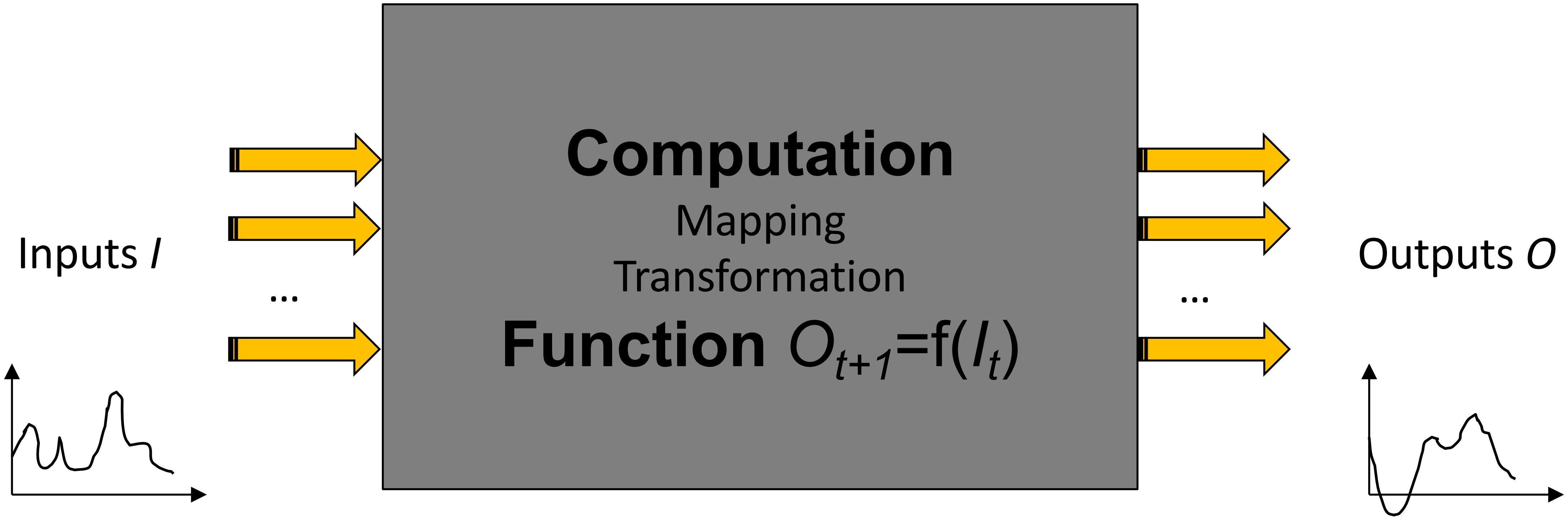
- Uses NNO artificial neurons + synapses
- Grows nodes when new classes appear
- Beats fixed static net: +212% MNIST, +250% CUB-200
- Resists catastrophic forgetting

MNIST + ECG
high accuracy from memcapacitor reservoir

+212% / +250%
MNIST / CUB-200 vs. static network

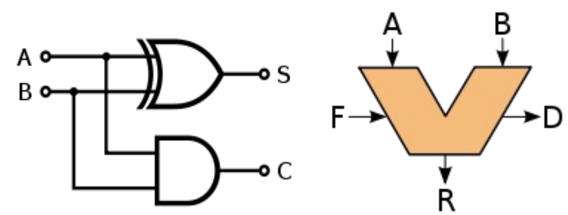
Hardware that reconfigures → networks that grow → no catastrophic forgetting.

What is computation?



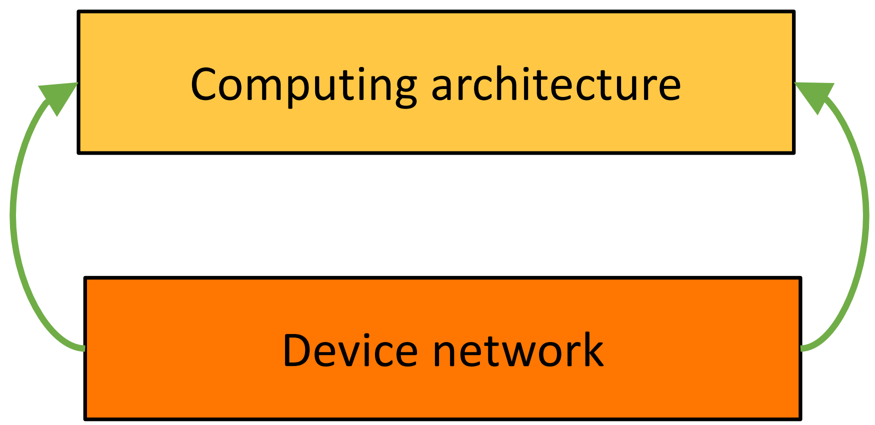
How can we “realize” this mapping?

Intrinsic vs designed computation

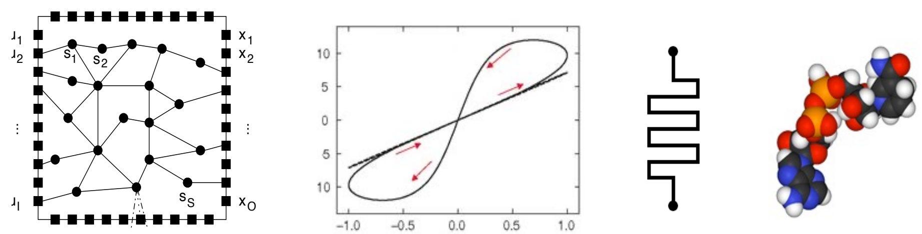


Desired computation

Intrinsic dynamics



How to obtain a desired computation?

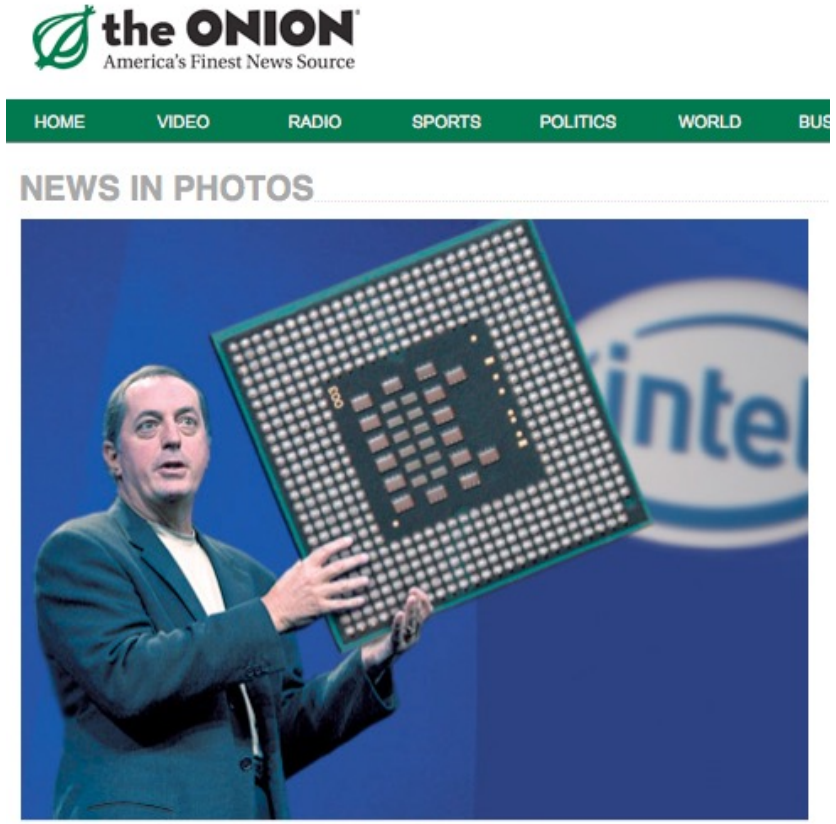


What is *Unconventional Computing* (UCOMP)?

There is no precise definition…

"[…] a computing scheme that today viewed as unconventional may well be so because its time hasn't come yet – or is already gone."
— Tommaso Toffoli

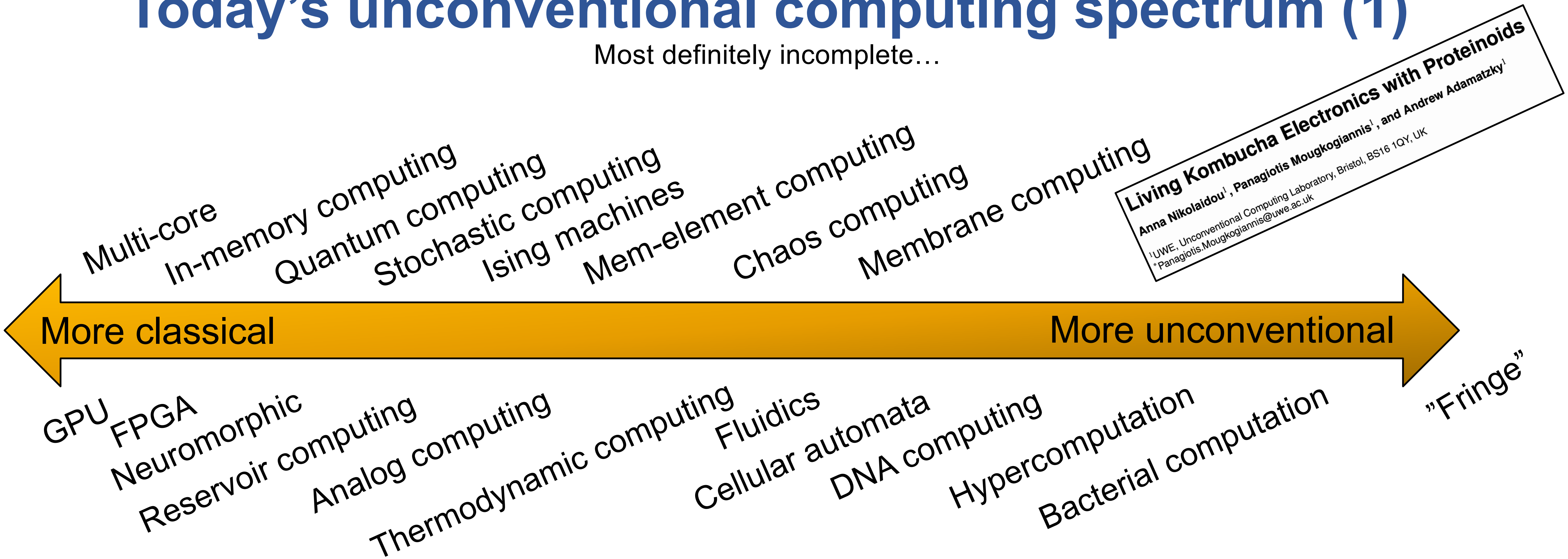
"Today, UCOMP is broad but (relatively) shallow, whilst CCOMP is narrow, but incredibly deep. What would computation look like if UCOMP were as deep as CCOMP, and there were an integrated theory combining all its aspects?"
— Susan Stepney



Intel Unveils Oversized Novelty Processor

Today’s unconventional computing spectrum (1)

Most definitely incomplete…



What should I “invest” in to reboot computing?

Materials – Devices & Circuits – Architectures – Software – Algorithms – Applications

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Today's unconventional computing topics spectrum (2)

Aims and Scope

The *International Journal of Unconventional Computing* offers the opportunity for rapid publication of theoretical and experimental results in non-classical computing. Specific topics include but are not limited to:

- physics of computation (e.g. conservative logic, thermodynamics of computation, reversible computing, quantum computing, collision-based computing with solitons, optical logic)
- chemical computing (e.g. implementation of logical functions in chemical systems, image processing and pattern recognition in reaction-diffusion chemical systems and networks of chemical reactors)
- bio-molecular computing (e.g. conformation based, information processing in molecular arrays, molecular memory)
- cellular automata as models of massively parallel computing
- complexity (e.g. computational complexity of non-standard computer architectures; theory of amorphous computing; artificial chemistry)
- logics of unconventional computing (e.g. logical systems derived from space-time behavior of natural systems; non-classical logics; logical reasoning in physical, chemical and biological systems)
- smart actuators (e.g. molecular machines incorporating information processing, intelligent arrays of actuators)
- novel hardware systems (e.g. cellular automata VLSIs, functional neural chips)
- mechanical computing (e.g. micromechanical encryption, computing in nanomachines, physical limits to mechanical computation).

Aims & Scope

npj Unconventional Computing considers all aspects of research applying unconventional approaches to improve computing efficiency and performance.

The journal covers a broad range of topics including but not limited to:

- Brain-inspired algorithms
- Quantum computing and quantum-inspired algorithms
- Molecular, and DNA computing
- Membrane computing and P systems
- Neuromorphic computing and brain-inspired architectures
- Amorphous computing and swarm robotics
- Cellular automata and complex systems
- Artificial life and evolutionary computation
- Chaos computing and dynamical systems
- Reversible and adiabatic computing
- Thermodynamic computing
- Probabilistic computing (p-bit computing)
- Superconducting computing
- Application of physical phenomena to computing
- Design, fabrication, and testing of unconventional hardware, physical substrates, and emerging technologies for computing

npj | unconventional computing

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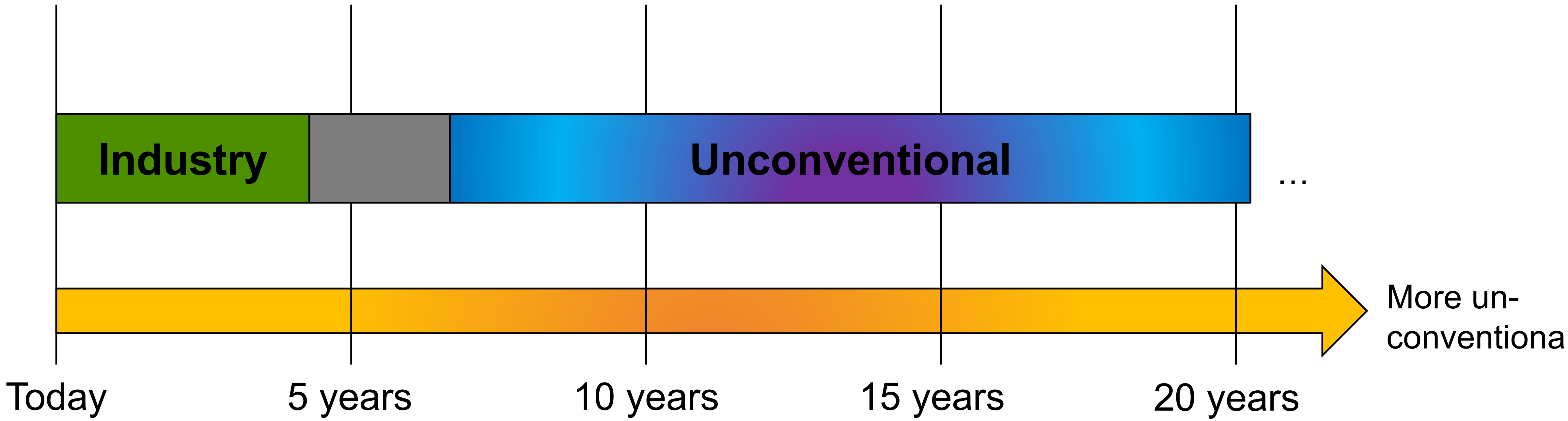
The unconventional computing time horizon



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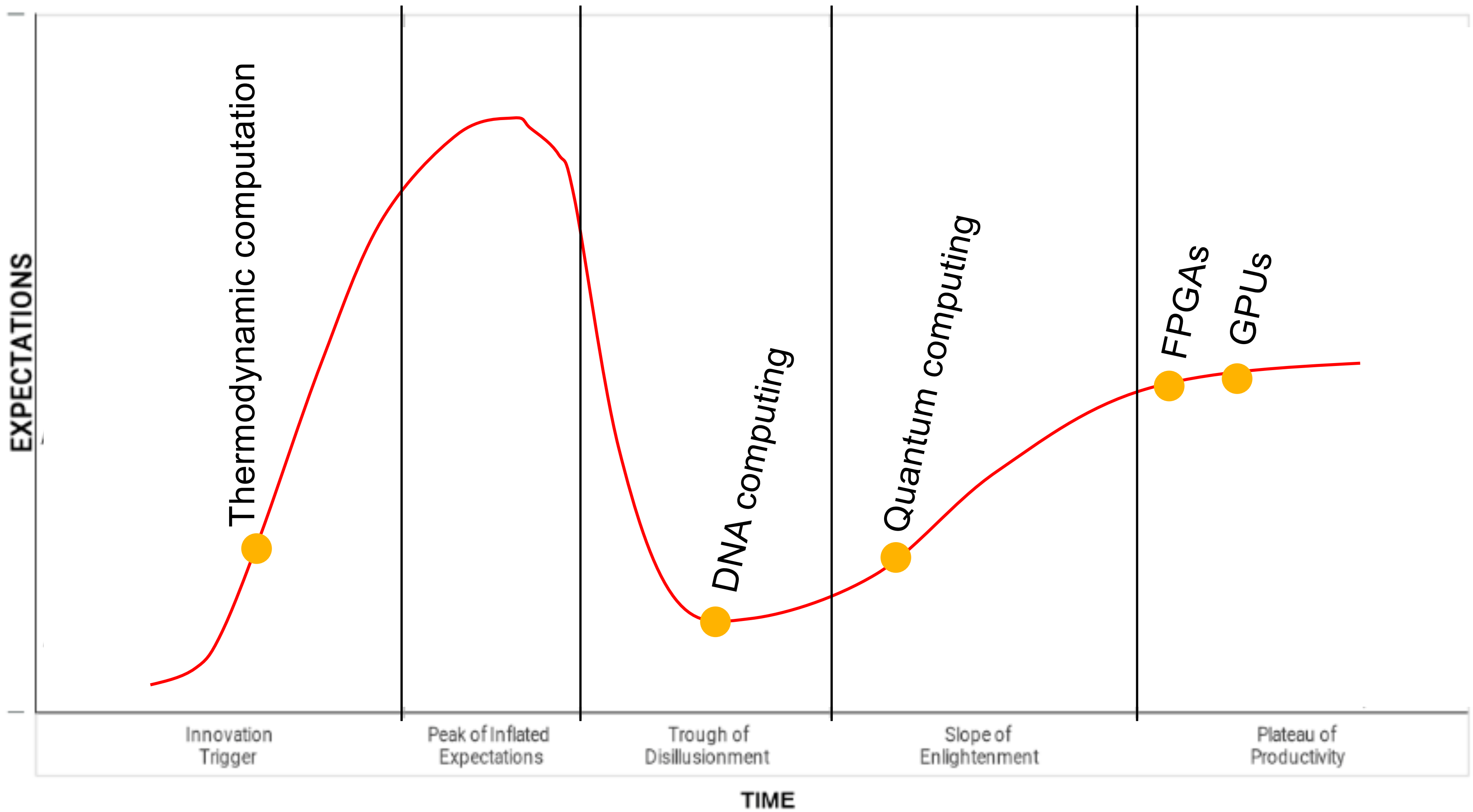
Hype or computation?



Gartner's hype cycle

UCOMP supporters 👍

- New technologies need time (and money) to mature.
- Current comparison with state-of-the-art are not fair comparisons.
- We need to start somewhere.
- Often on shoestring budgets only.

UCOMP critics 👎

- No useful paradigms that can compete with conventional approaches.
- The real, practical challenges to be solved are always kept comfortably far away.
- Proving that we can compute a NAND function with a self-assembled nanowire network, for example, doesn't move the needle.

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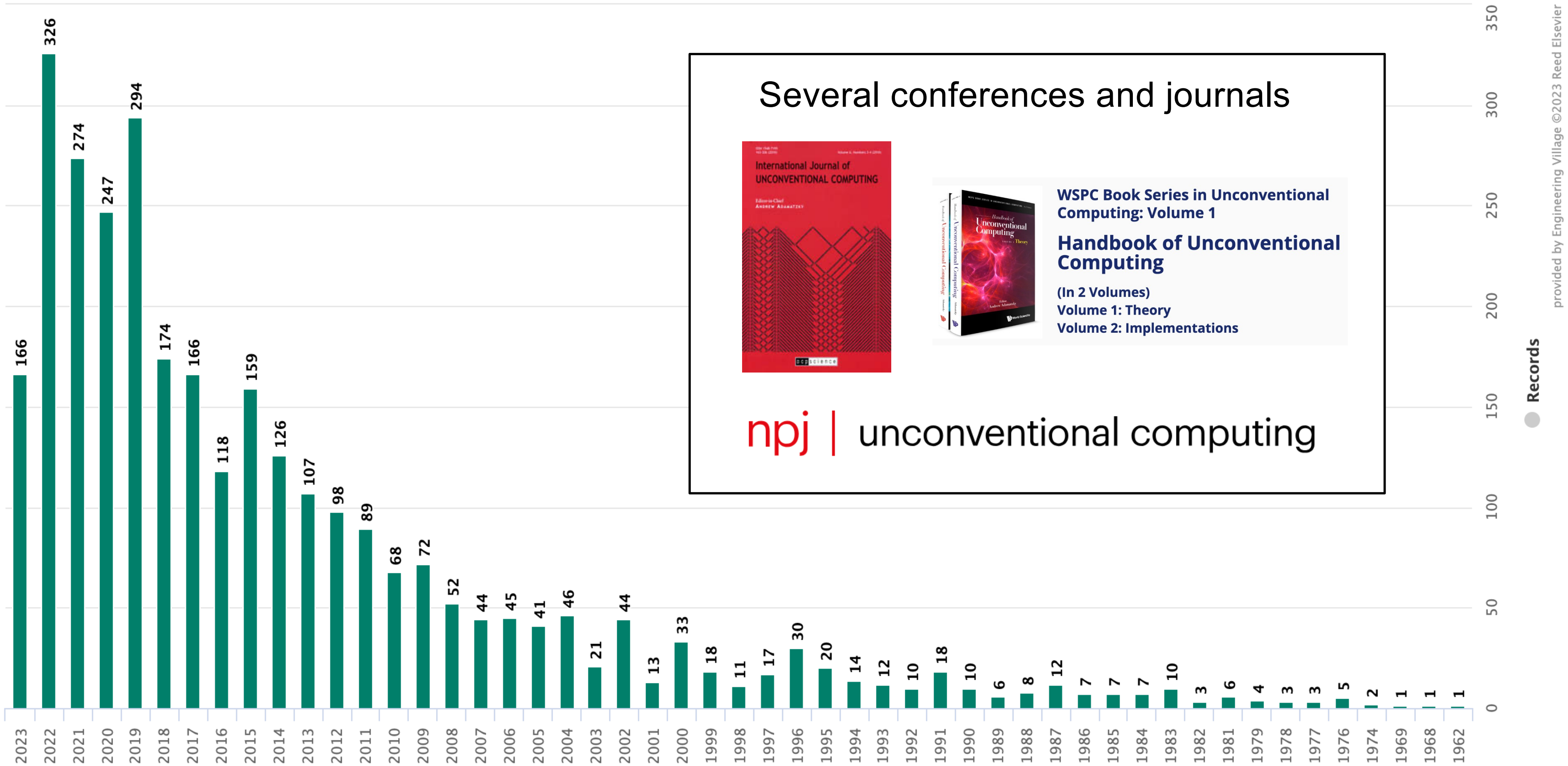
Evolution of the popularity of "unconventional computing"

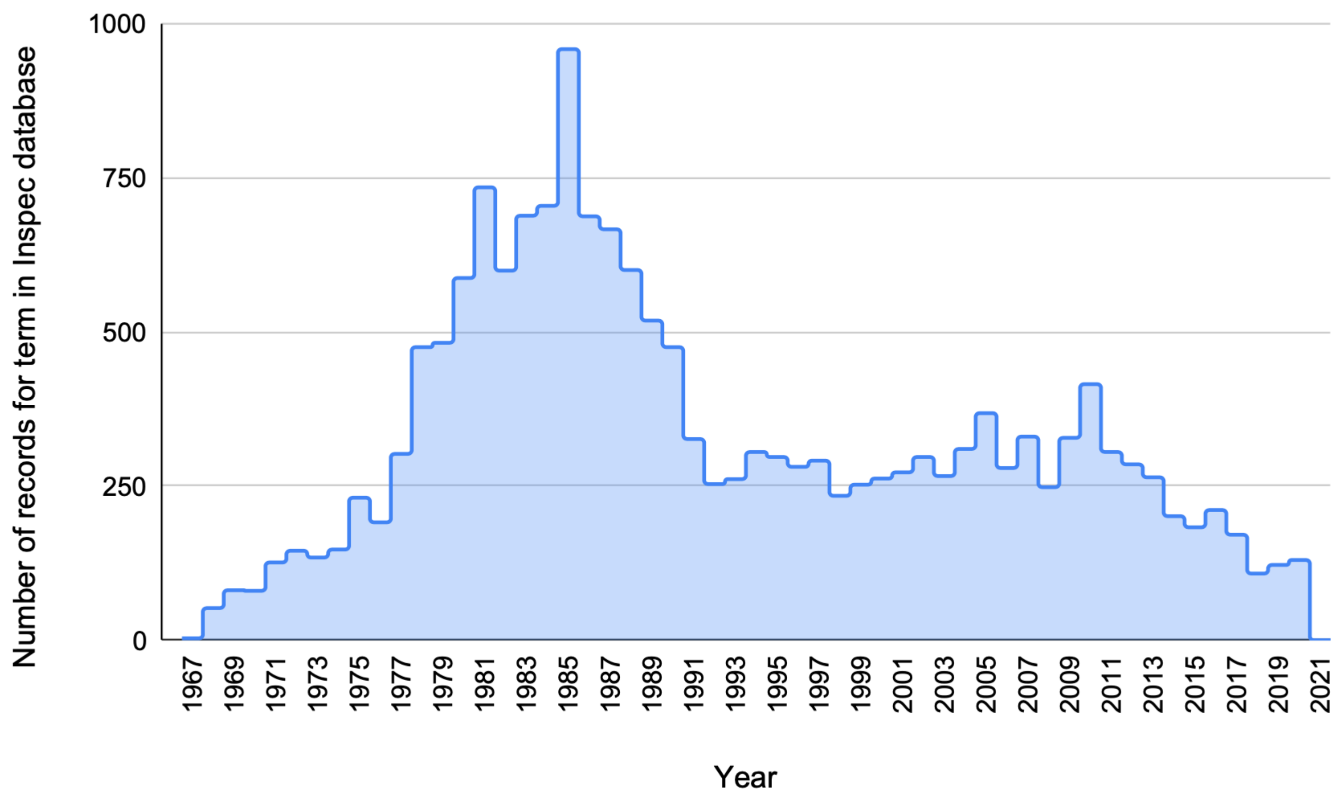
Time



Source: Engineering Village, Inspec

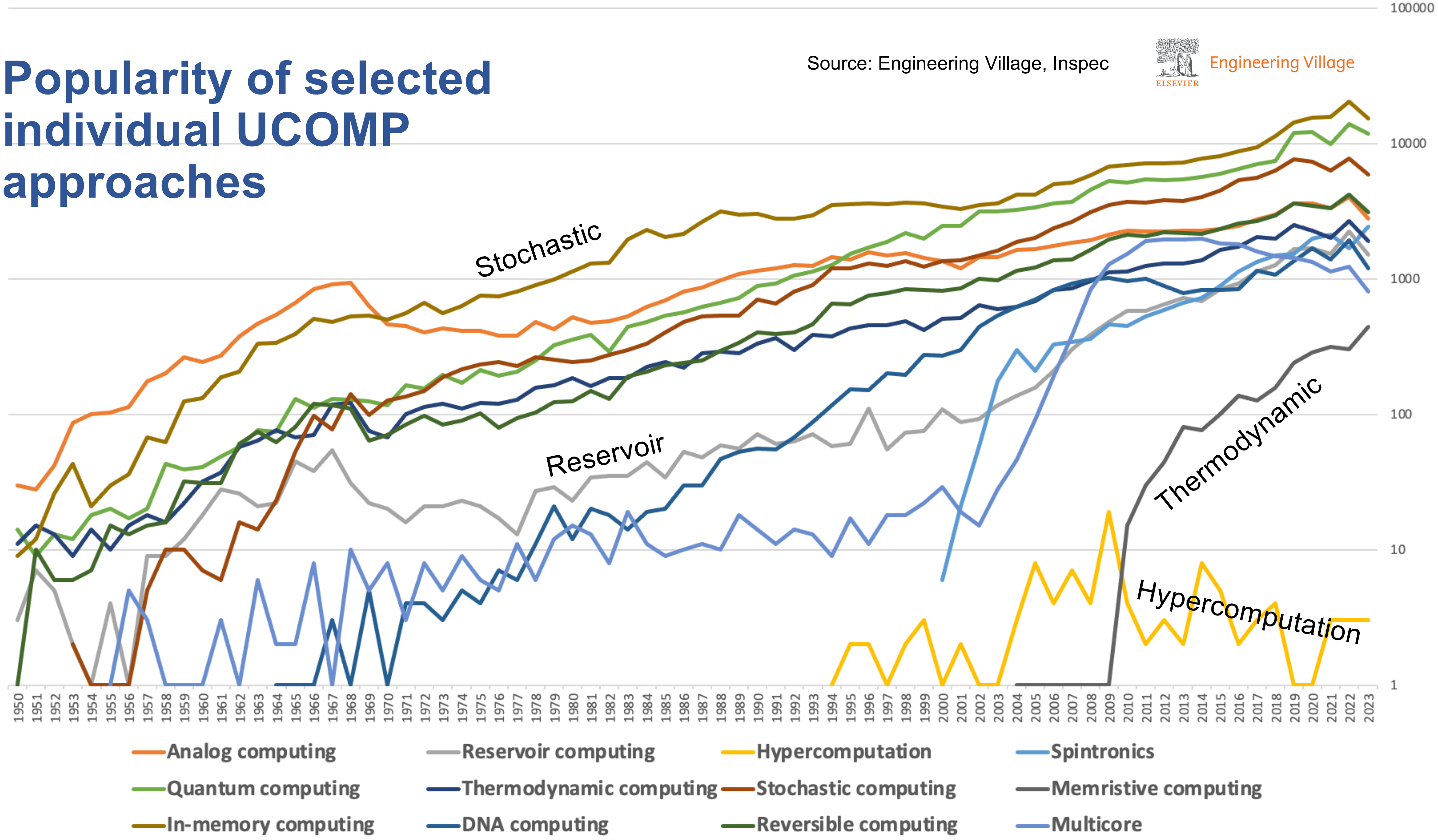
Engineering Village

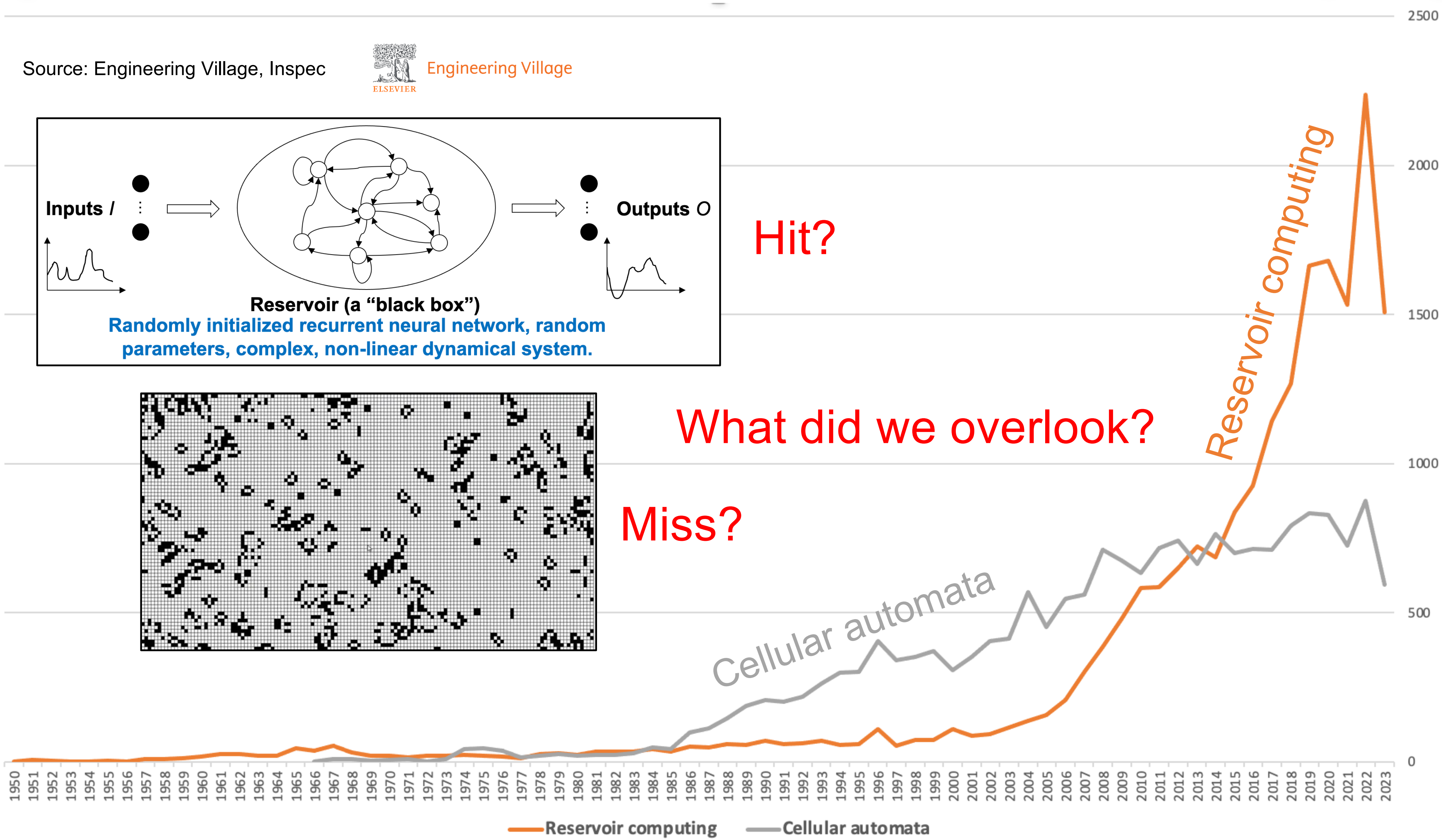
The decline of traditional computer architecture



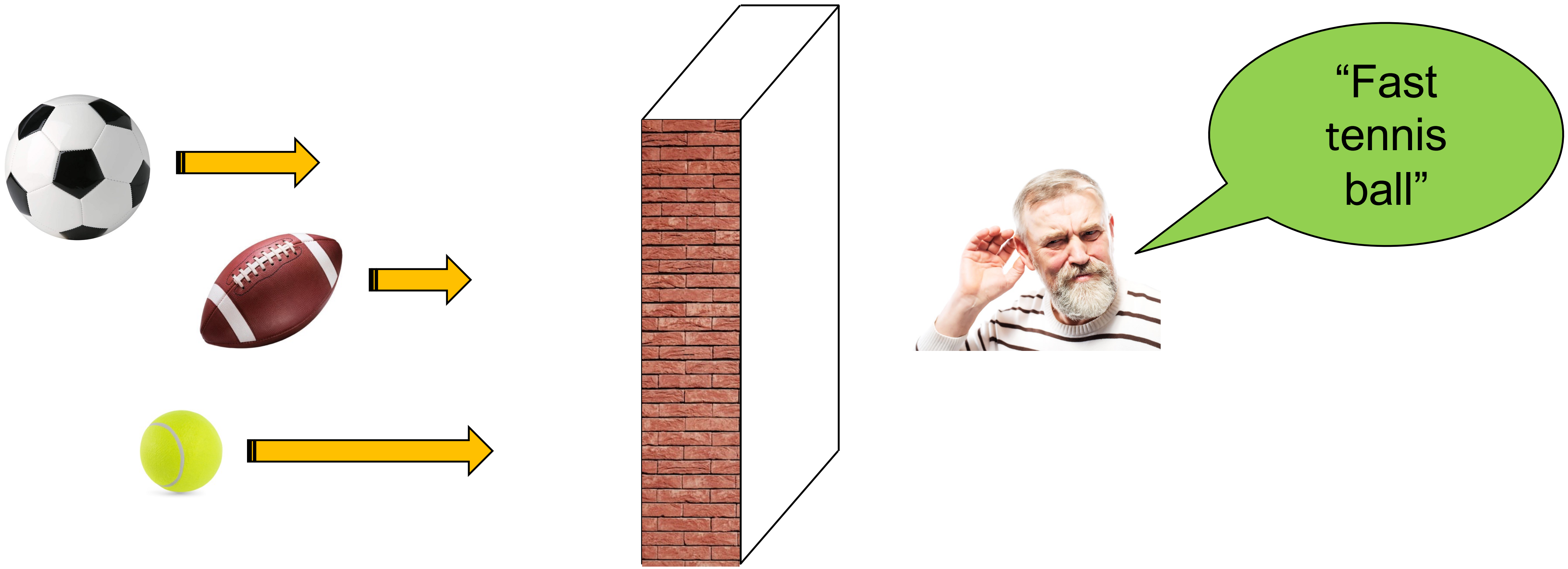
The number of records over the years that were found for the “**computer architecture**” thesaurus term. The field had its peak around 1985 and has since been in decline. Data source: Elsevier Engineering Village thesaurus.

Popularity of selected individual UCOMP approaches

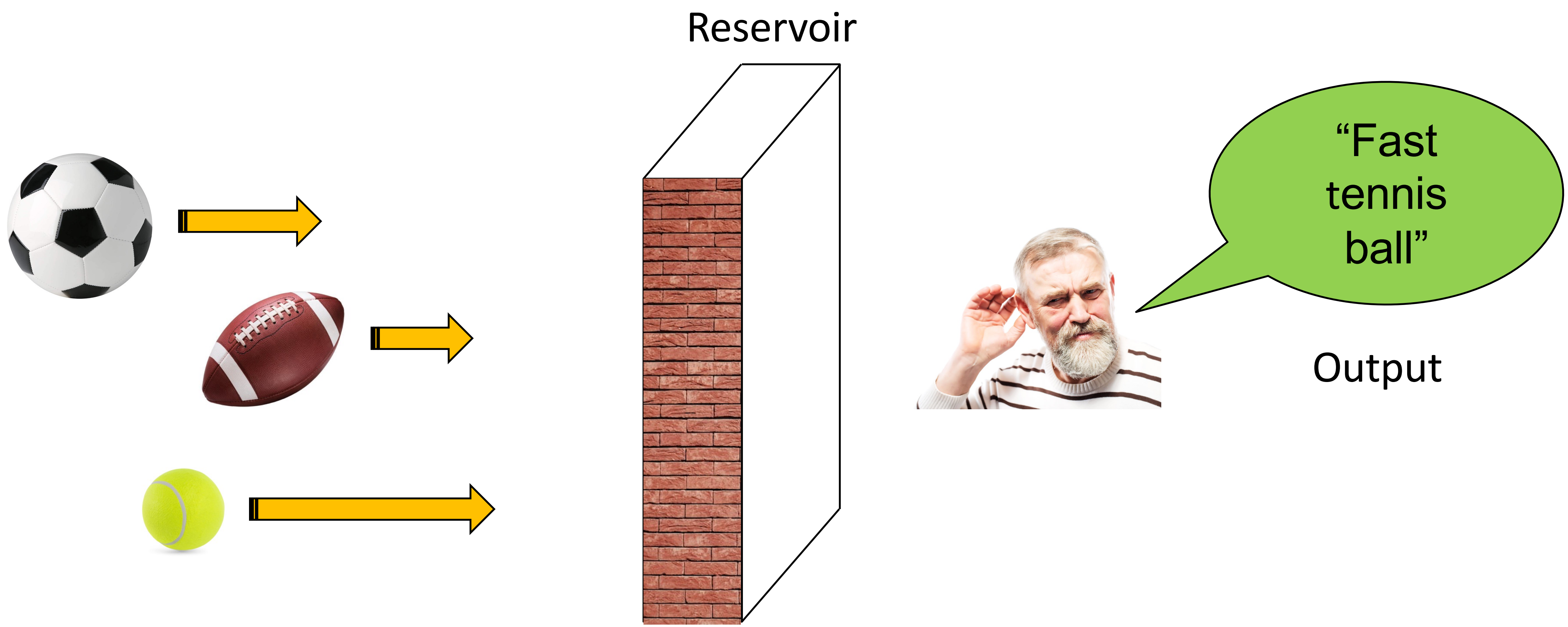



Source: Engineering Village, Inspec
Engineering Village
ELSEVIER
Inputs *I*
Outputs *O*
Reservoir (a "black box")
Randomly initialized recurrent neural network, random parameters, complex, non-linear dynamical system.
Hit?
What did we overlook?
Miss?
Reservoir computing
Cellular automata
2500
2000
1500
1000
500
0
Reservoir computing
Cellular automata

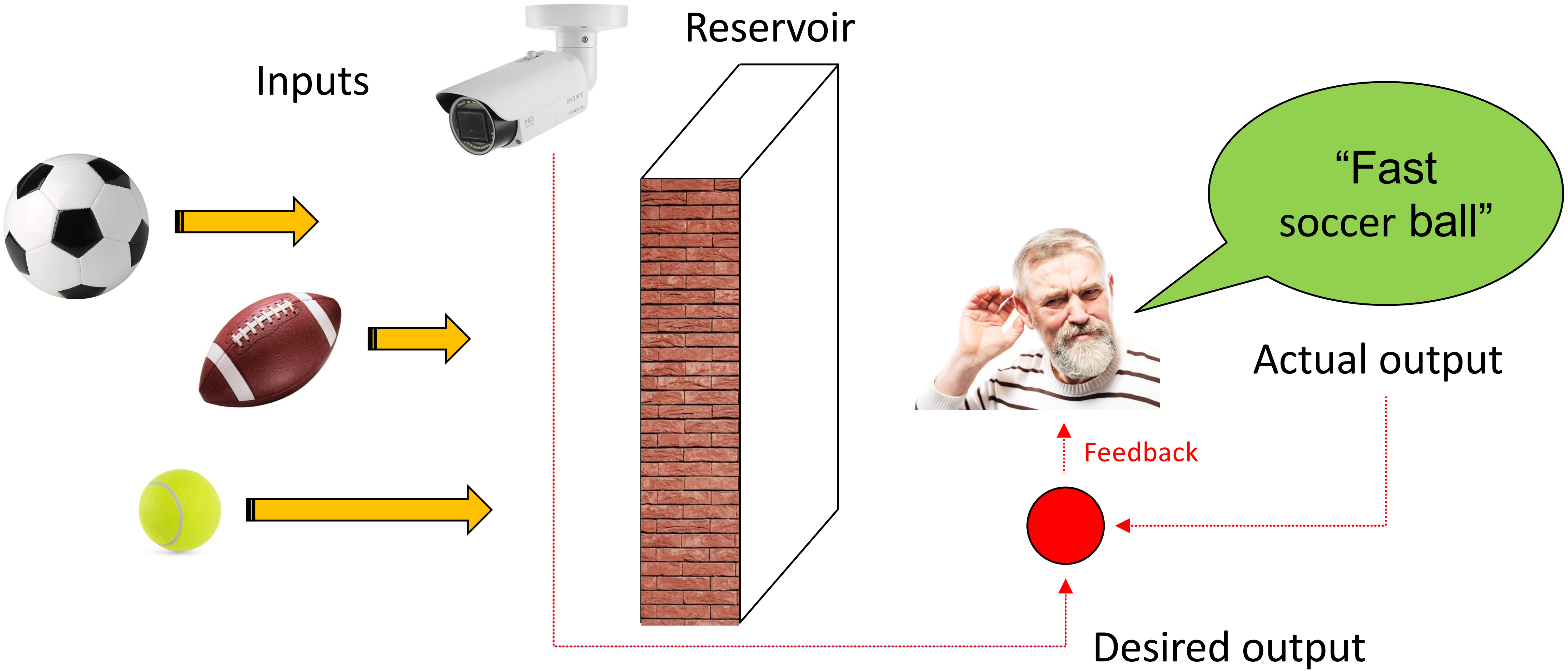
Reservoir computing analogy (1)



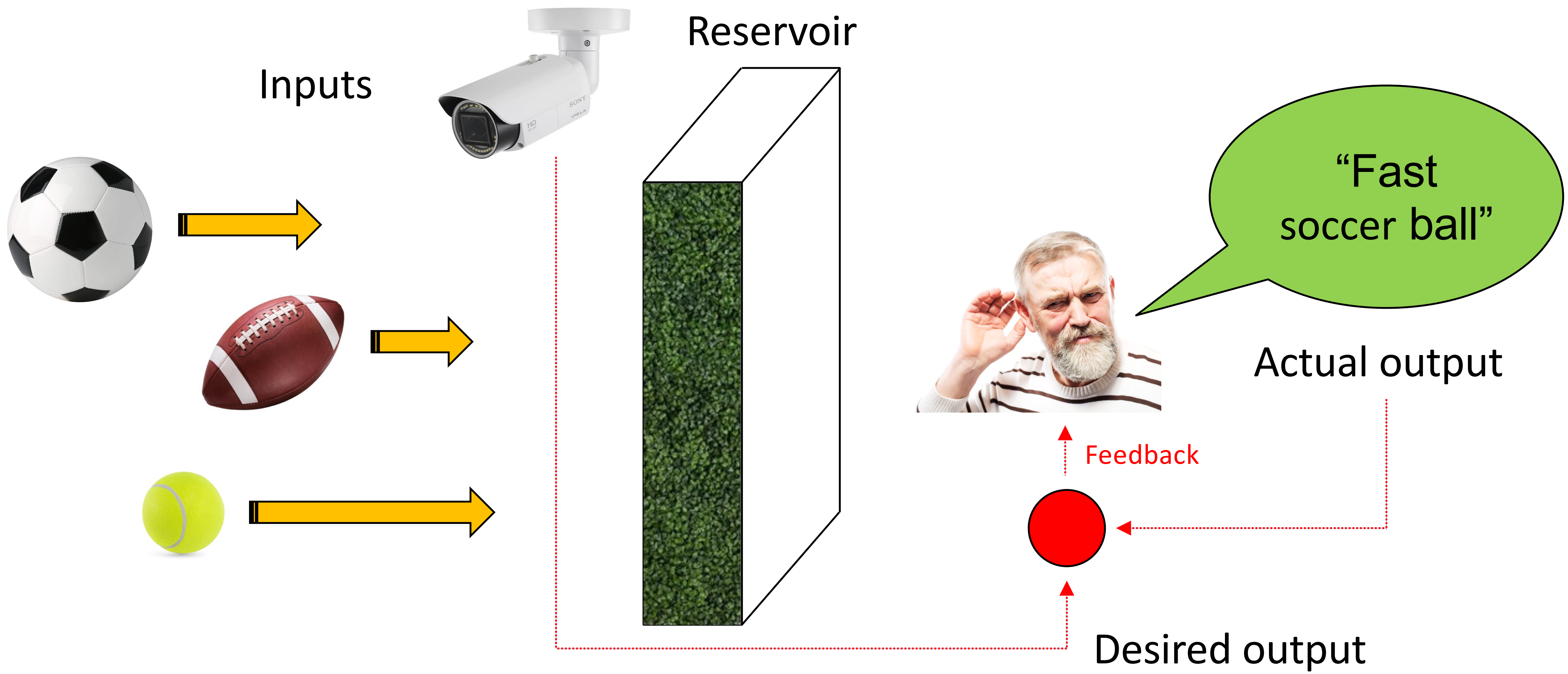
Reservoir computing analogy (2)



Reservoir computing analogy (3)



Reservoir computing analogy (4)



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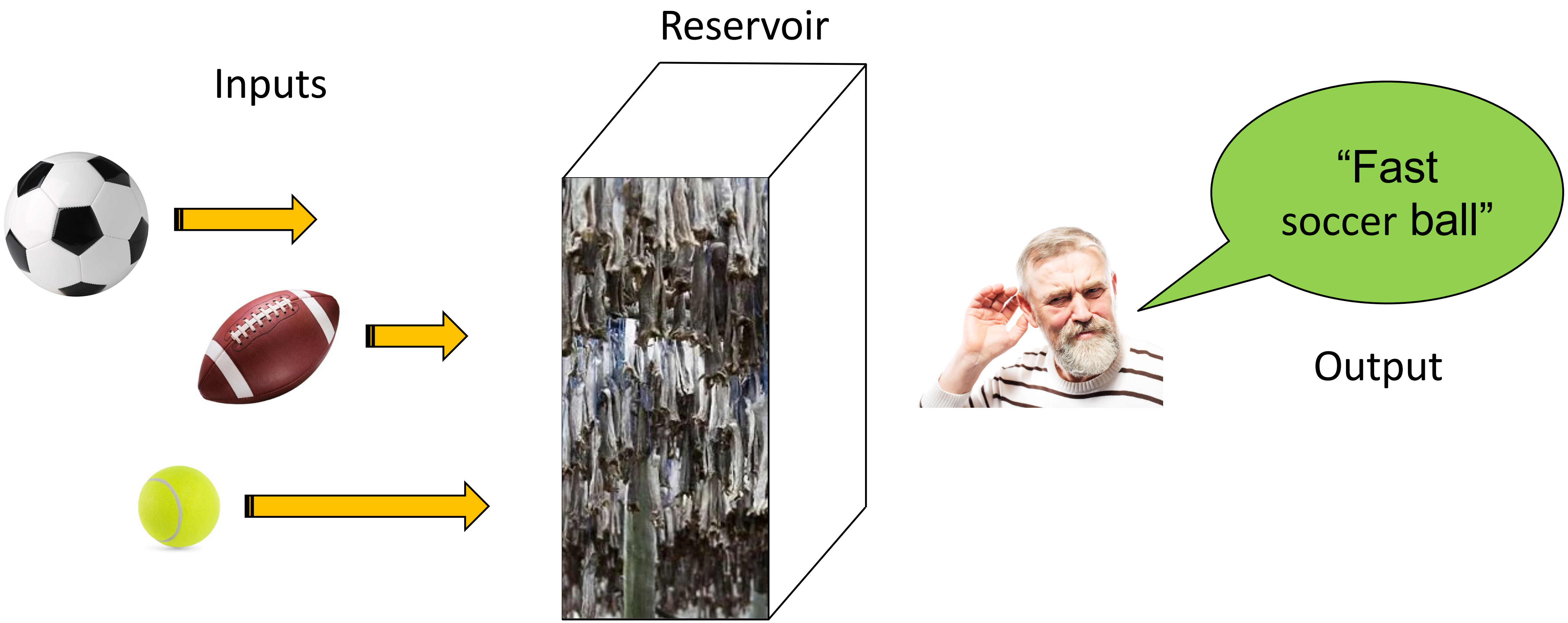
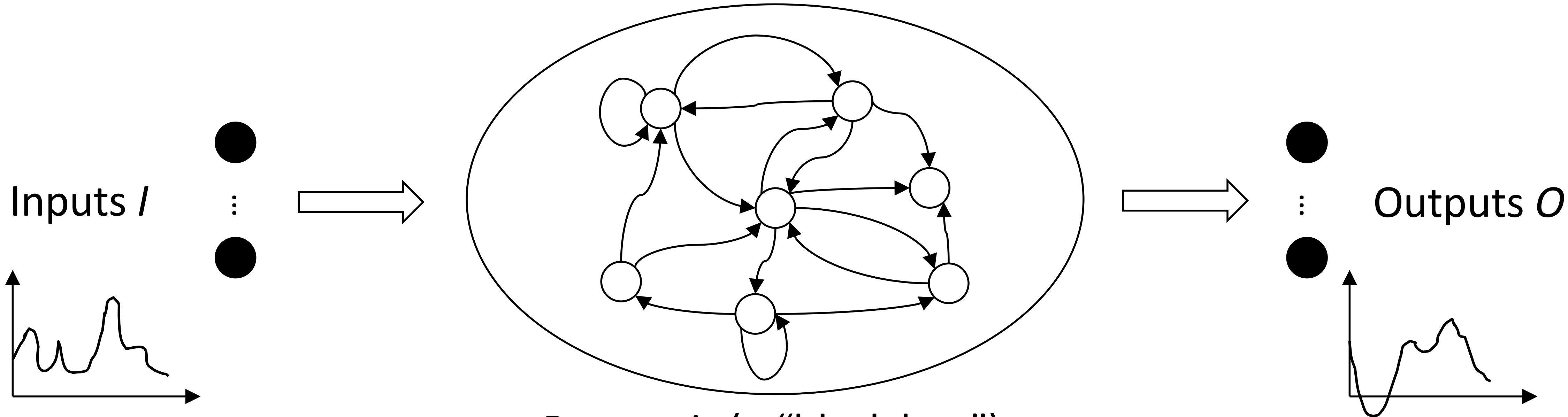
Reservoir computing analogy (5)



Components of a reservoir computer (1)

Reservoir computing addresses issues of traditional recurrent neural networks: lower learning complexity, less weights to train, harnessing intrinsic dynamics of a dynamical system, etc.



Reservoir (a "black box")

Randomly initialized recurrent neural network, random parameters, complex, non-linear dynamical system.

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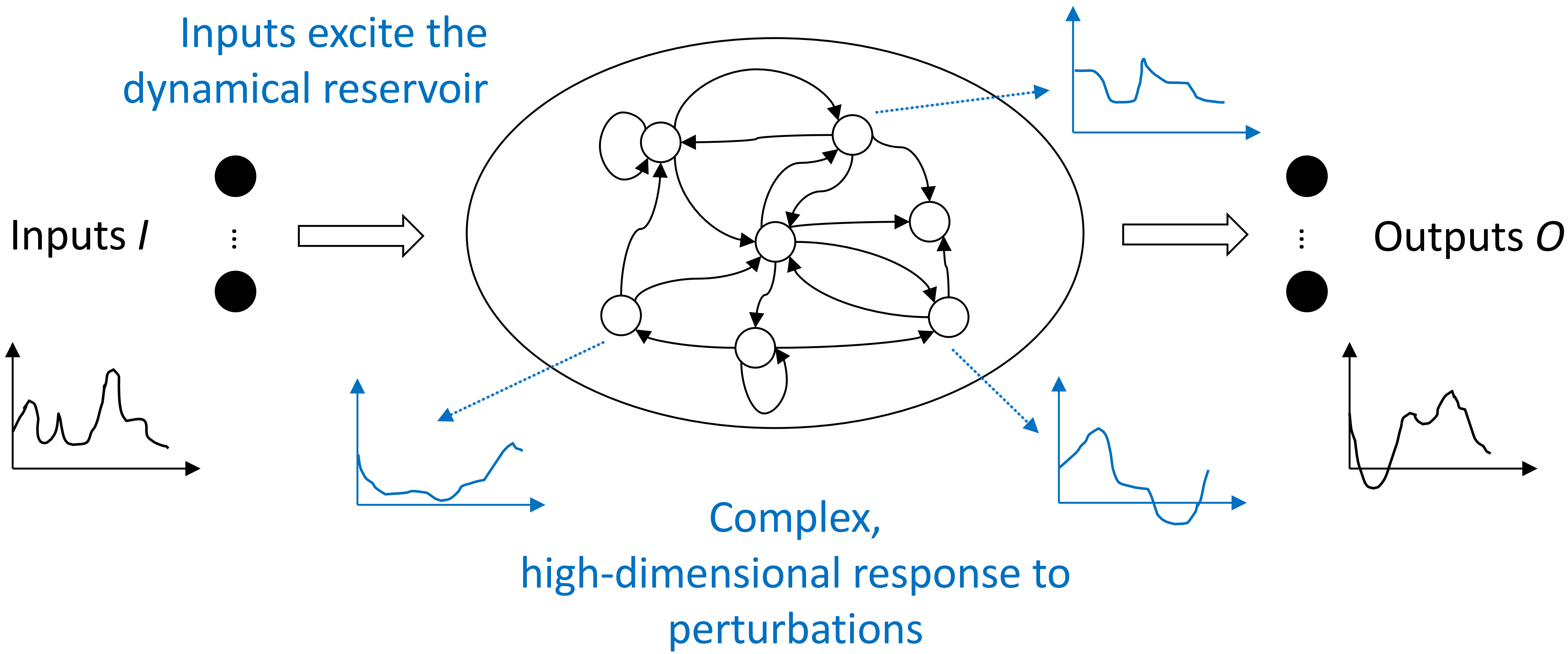
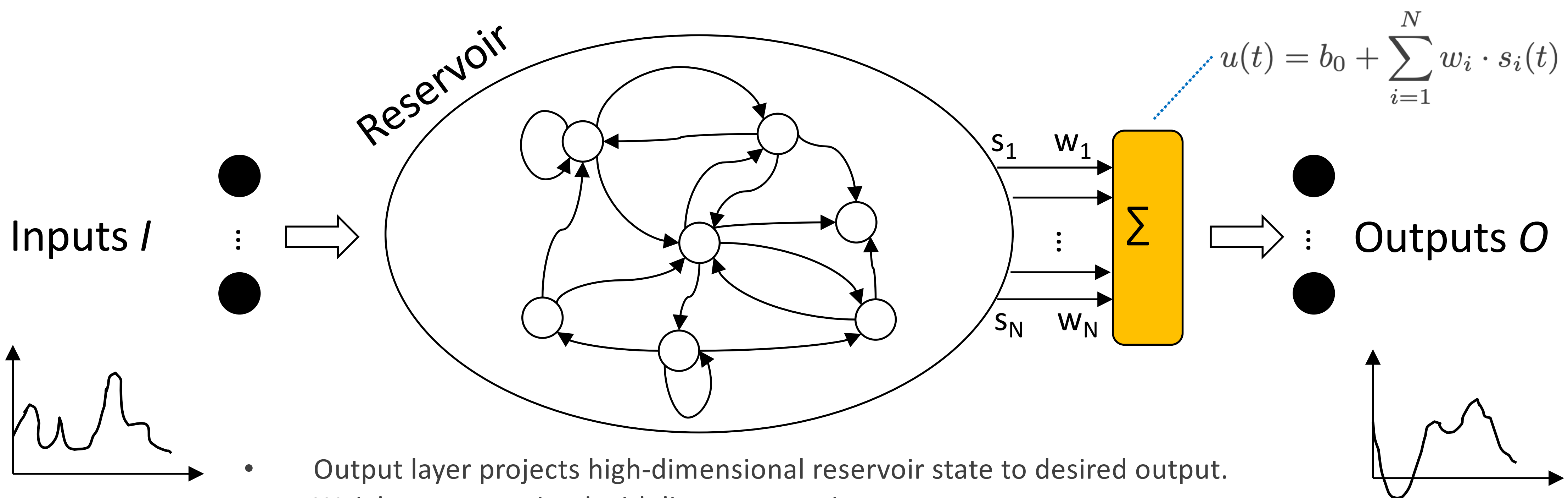
Components of a reservoir computer (2)

Reservoir computing addresses issues of traditional recurrent neural networks: lower learning complexity, less weights to train, harnessing intrinsic dynamics of a dynamical system, etc.



Components of a reservoir computer (3)

Reservoir computing addresses issues of traditional recurrent neural networks: lower learning complexity, less weights to train, harnessing intrinsic dynamics of a dynamical system, etc.



- Output layer projects high-dimensional reservoir state to desired output.
- Weights w_i are trained with linear regression.
- No training occurs in reservoir.

What makes a good reservoir?

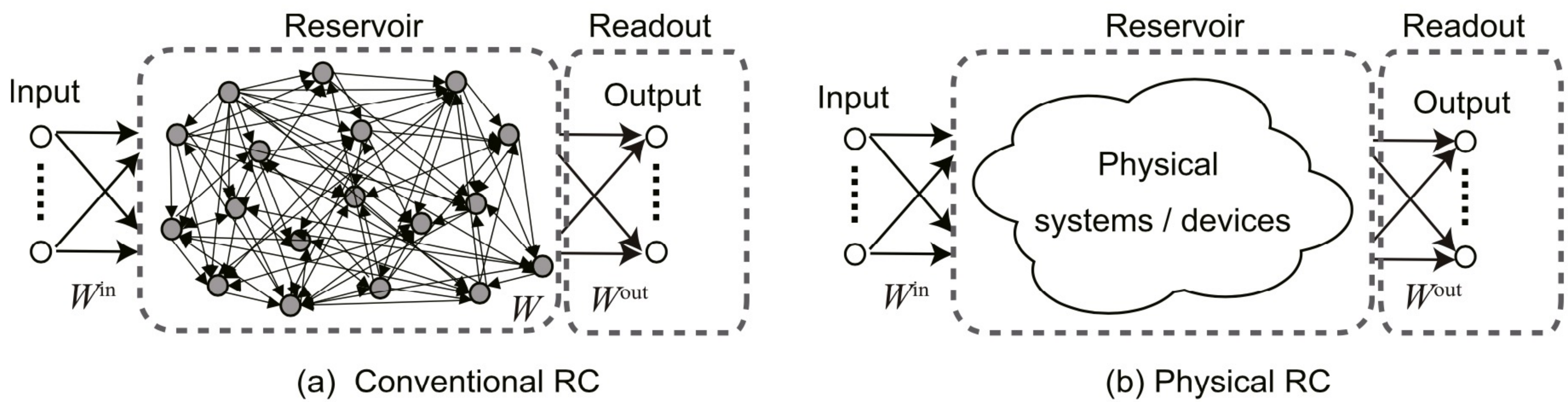
Intuitively:
- Complex-enough, non-linear dynamics
- Large enough

More formally, reservoirs must have following two properties:
- Separation property or Kernel quality
- Fading memory or Echo State property (ESP)

Lukoševičius, A Practical Guide to Applying Echo State Networks, 2012, https://doi.org/10.1007/978-3-642-35289-8_36

What is material and Physical Reservoir Computing (PRC)?

A physical reservoir computer is a system in which the reservoir is realized with a physical system.



Source: Tanaka et al., Recent advances in physical reservoir computing: A review, 2019, https://doi.org/10.1016/j.neunet.2019.03.005

Questions of interest for Physical Reservoir Computing (PRC)

- How much control does on have over the fabrication process?
- Reproducibility of input-output mapping?
- How to interface with a material/device?
- Memory capacity?
- Signal attenuation/amplification?
- Cycle-to-cycle variation?
- Device-to-device variation?
- In-situ or ex-situ training? The challenging part tends to be the *in materio* implementation of the readout and training.
- What pre- and post-processing is necessary? What will it "cost?"
- …

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Pattern recognition in a bucket (1)

4 electric motors perturb water surface

320x240 pixels, 5 frames webcam films inference patters

Sobel edge detection mask that was suitably thresholded to remove noise and averaged to produce a 32x24 grid. The cells of this matrix were used as the input values for 50 perceptrons in parallel

Overhead projector

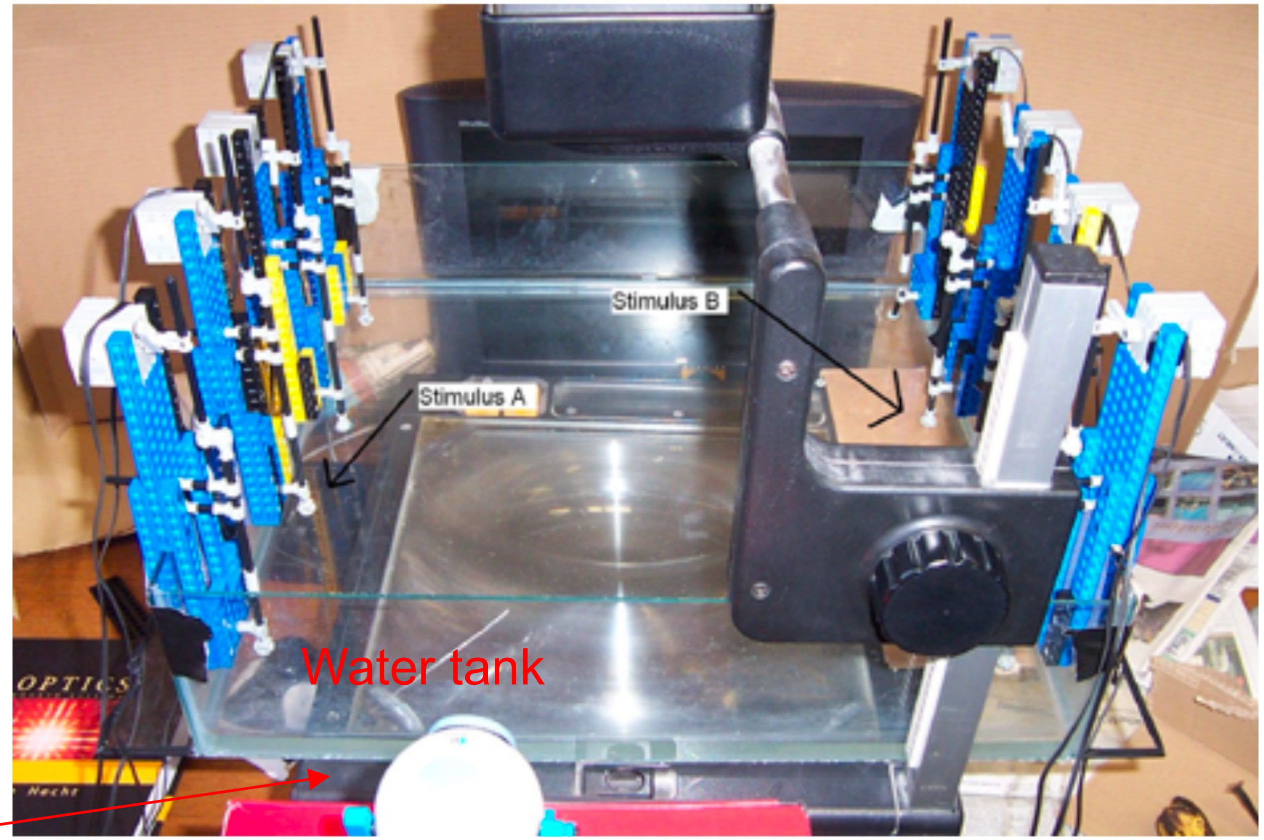
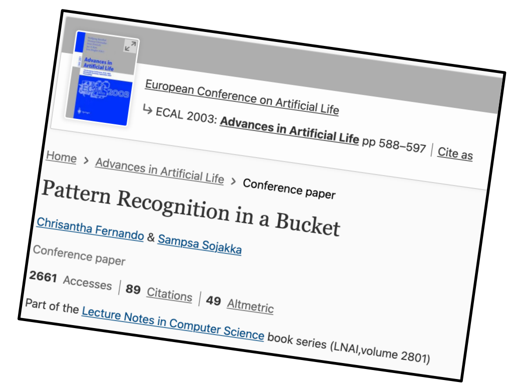


Fig. 1. The Liquid Brain.

- XOR task
- Speech recognition ("zero"/"one" task)
  - 25% mistakes with single perceptron
  - 1.5% mistakes when passed through water.



European Conference on Artificial Life

ECAL 2003: **Advances in Artificial Life** pp 588–597 | Cite as

Home > Advances in Artificial Life > Conference paper

Pattern Recognition in a Bucket

Chrisantha Fernando & Sampsa Sojakka

Conference paper

2661 Accesses | **89** Citations | **49** Altmetric

Part of the Lecture Notes in Computer Science book series (LNAI,volume 2801)

DOI: 10.1007/978-3-540-39432-7_63

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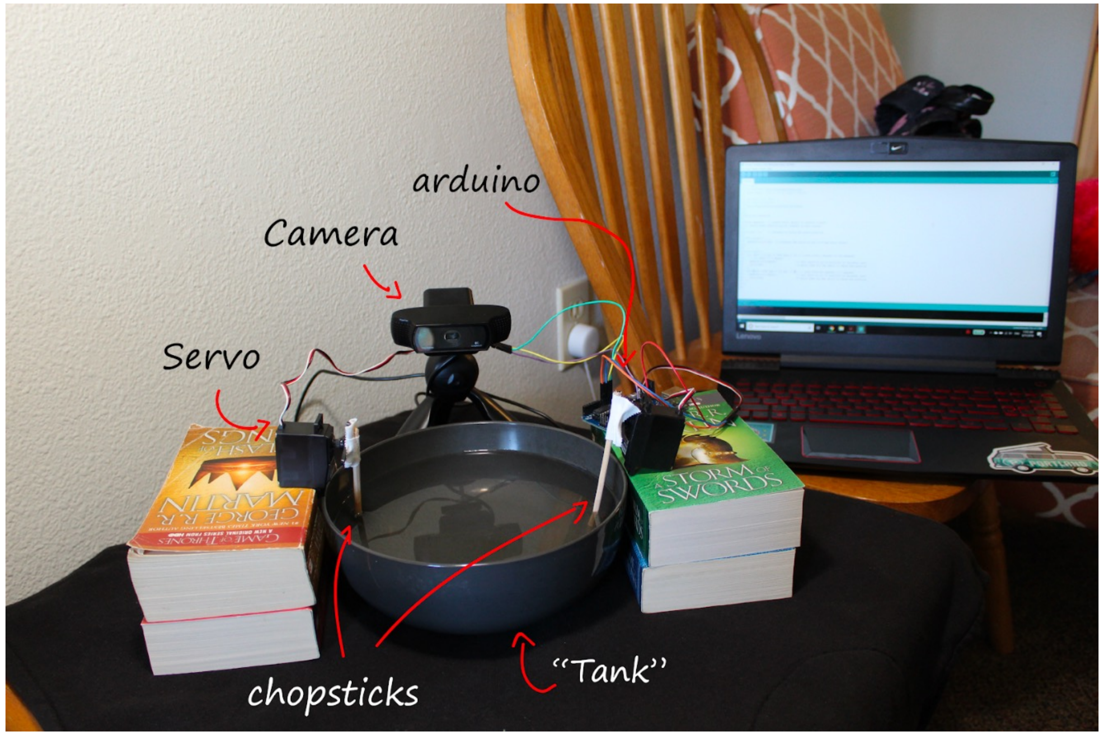
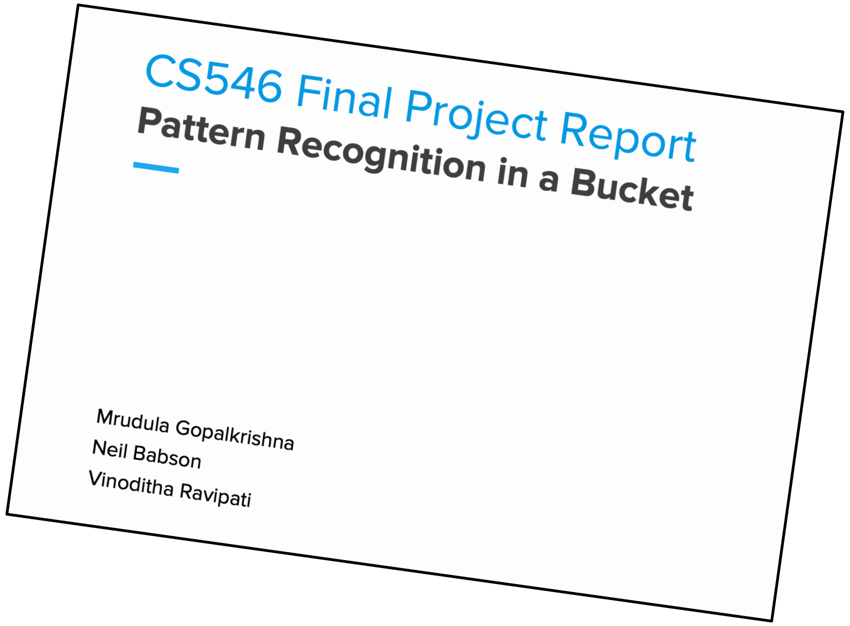
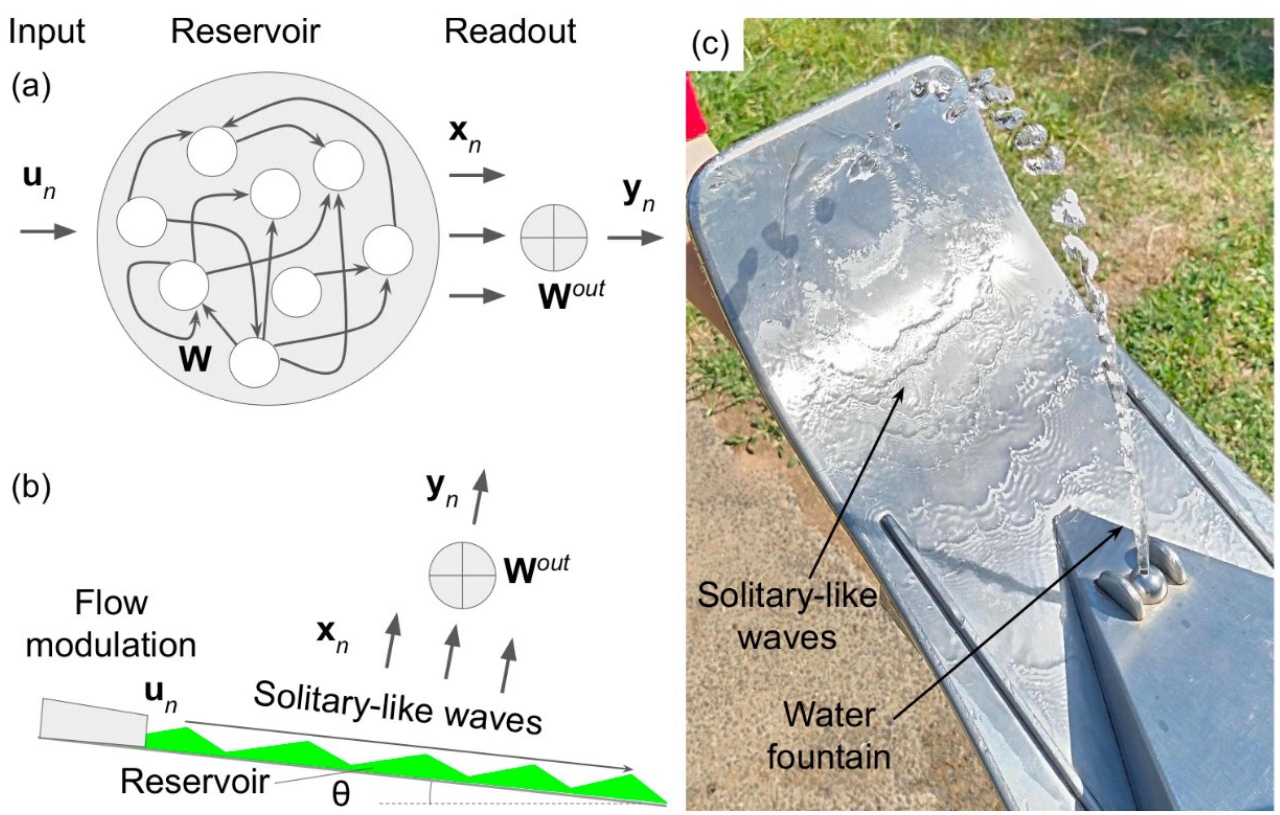
Pattern recognition in a bucket (2)



Fernando and Sajakka experiment replication.

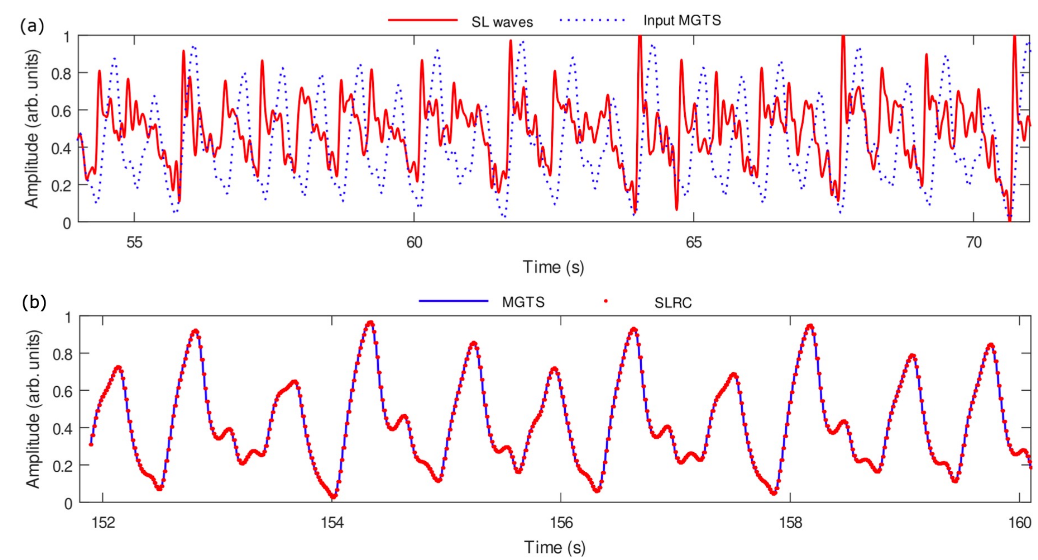


RC based on solitary-like waves dynamics of film flows



Maksymov & Pototsky, Reservoir computing based on solitary-like waves dynamics of film flows: a proof of concept, 2023, https://arxiv.org/abs/2303.01801

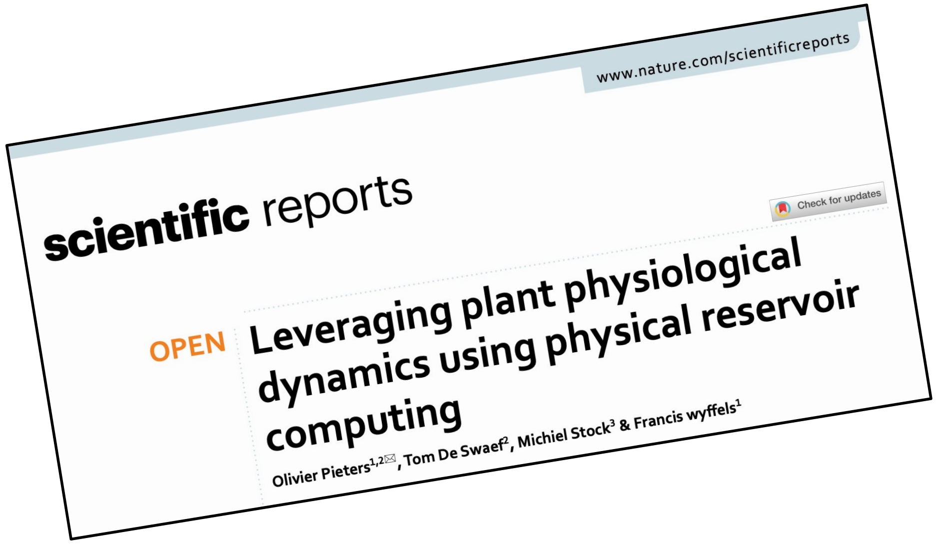
Mackey-Glass time series task



An electric signal modulates the pump flow: ’1’ states are implemented as a brief change in the flow rate that results in the generation of one.

Camera used for outputs. Output layer in software.

Plant reservoir computing


www.nature.com/scientificreports

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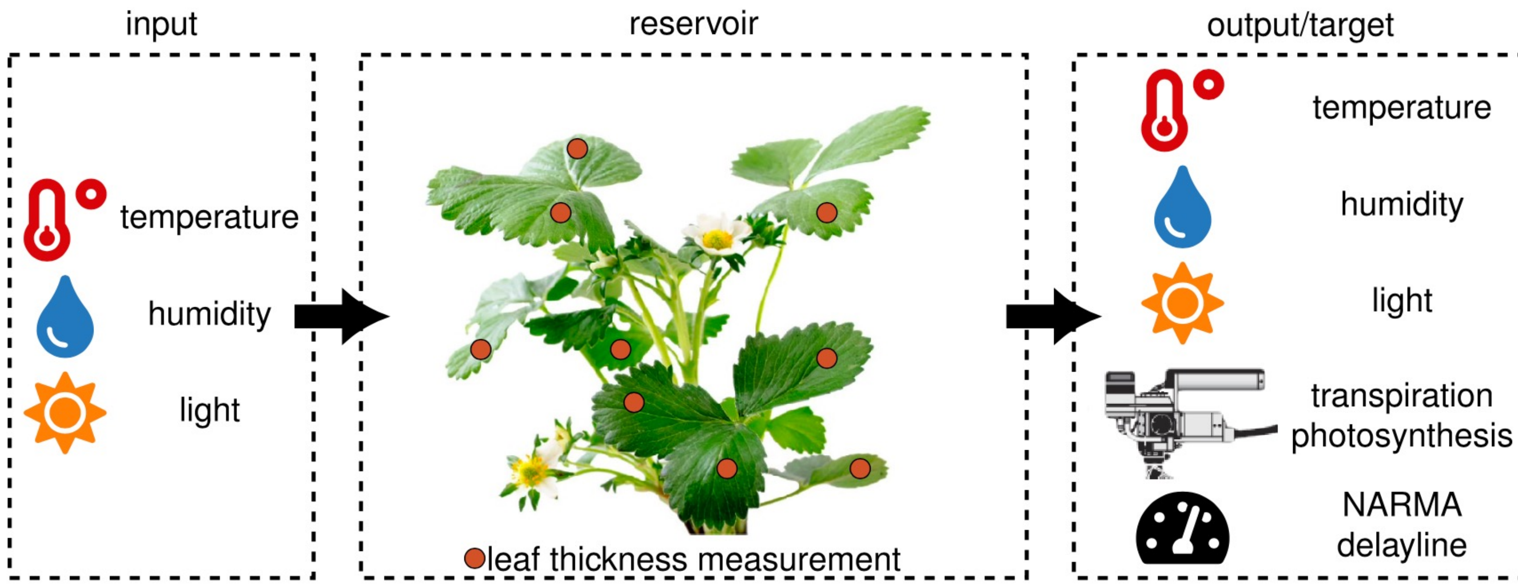
Leveraging plant physiological dynamics using physical reservoir computing

Olivier Pieters[1,2], Tom De Swaef[2], Michiel Stock[3] & Francis wyffels[1]



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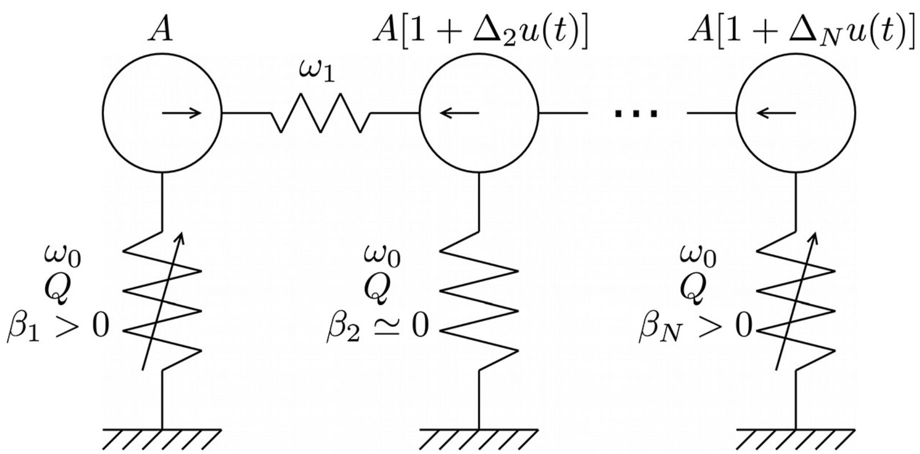
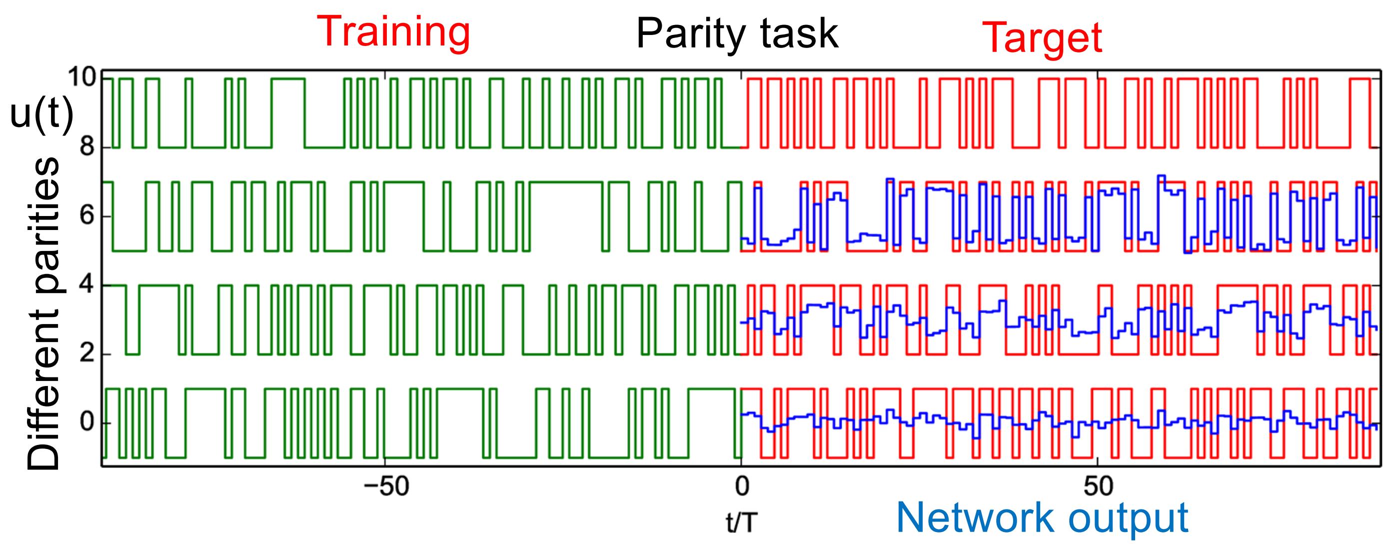
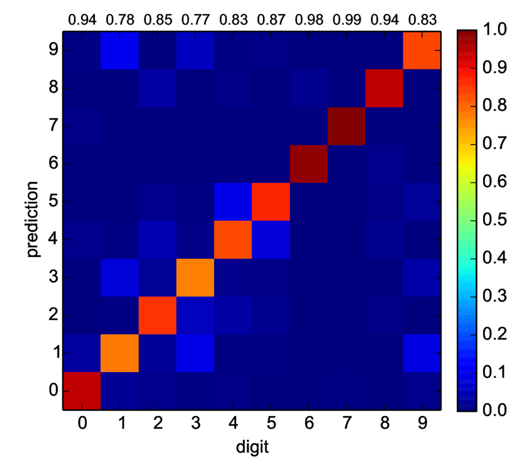
Computing with networks of nonlinear mechanical oscillators

Anharmonic = not simply harmonic
Springs with damping. Random couplings.





N= 400 oscillators
Parity + spoken words tasks
Numerical simulations only.



Coulombe, Computing with networks of nonlinear mechanical oscillators, 2017, https://doi.org/10.1371/journal.pone.0178663

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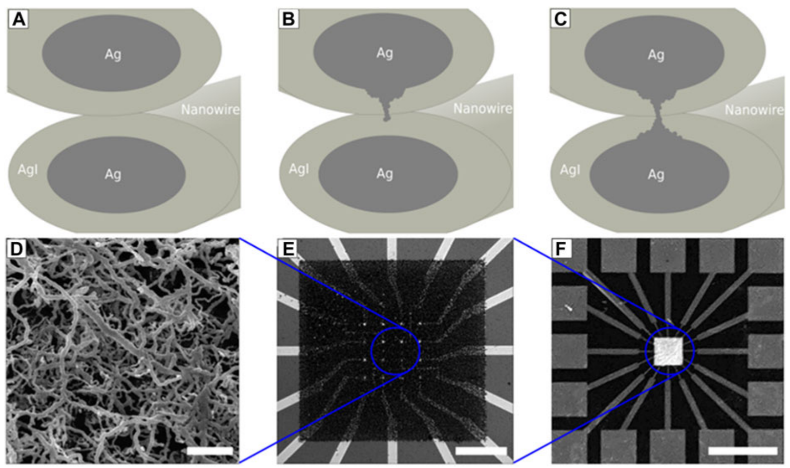
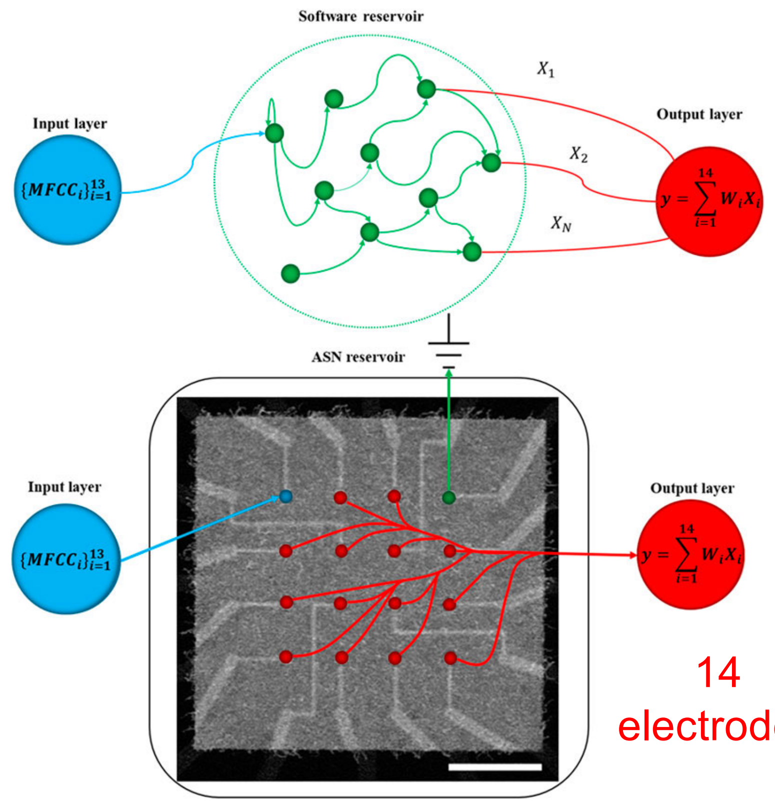
Reservoirs with atomic switch networks

Silver iodide (AgI) junctions

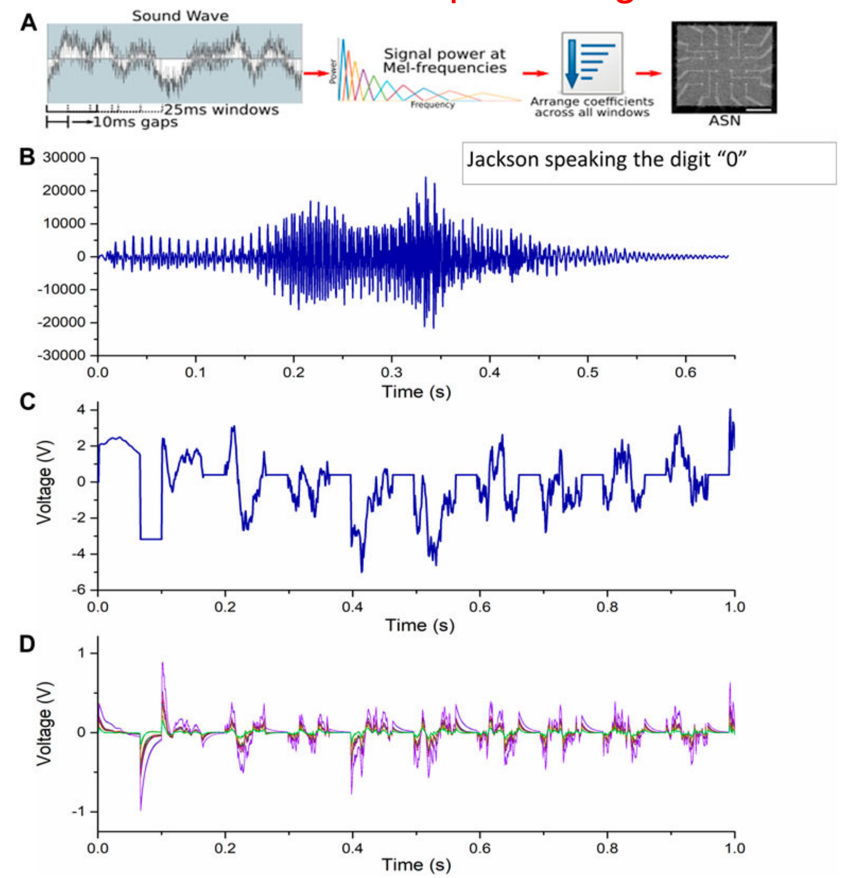


Self-organizing systems



14 electrodes

Spoken digits task



Lilak, Woods, Scharnhorst K, Dunham, Teuscher, Stieg, Gimzewski, Spoken Digit Classification by In-Materio Reservoir Computing With Neuromorphic Atomic Switch Networks, 2021, http://doi.org10.3389/fnano.2021.675792

Dynamic memristors for temporal information processing

Experimental demonstration of a memristor-based RC system using dynamic memristor devices that offer internal, short-term memory effects. 88 memristors.

Fig. 1

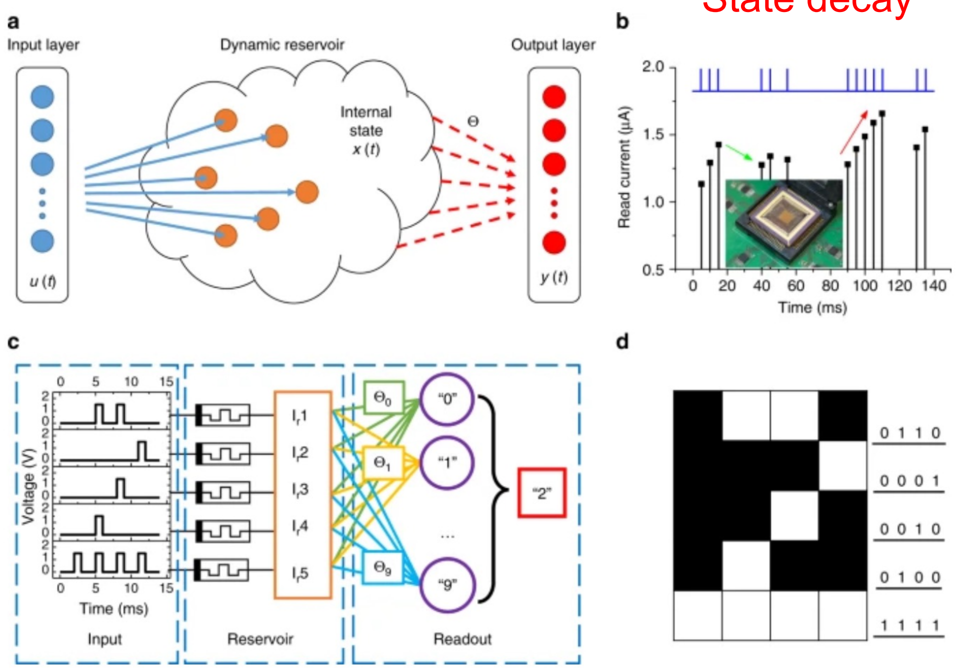
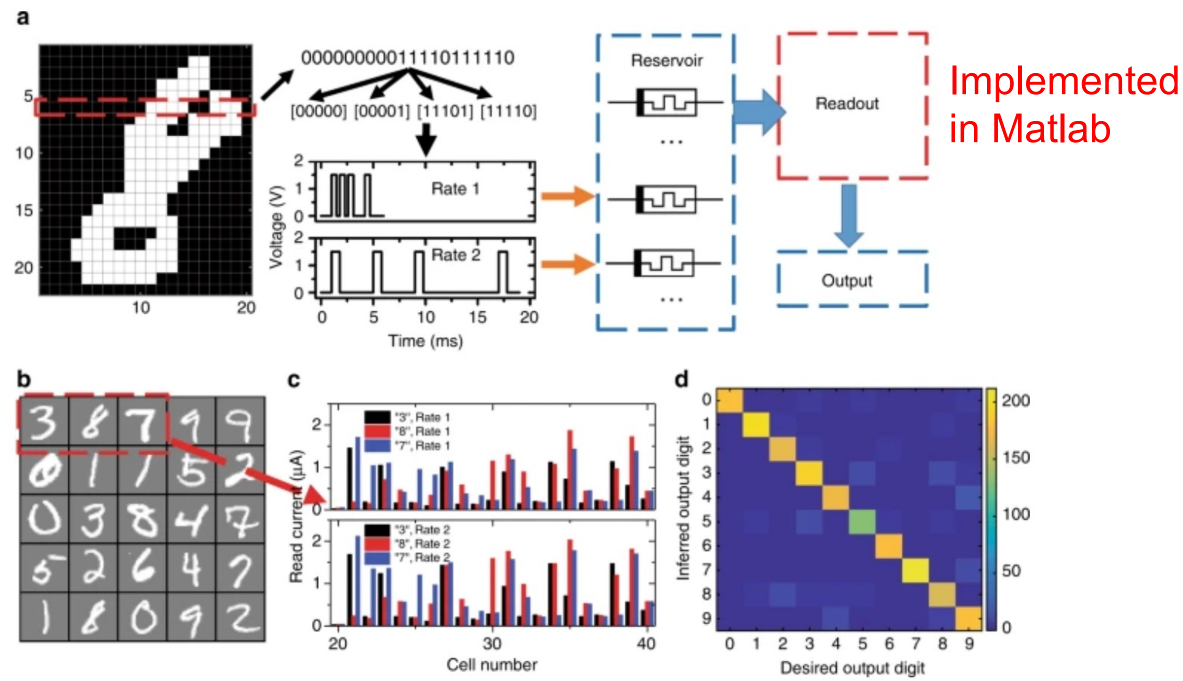


Fig. 4



Source: Du et al. Reservoir computing using dynamic memristors for temporal information processing. Nat Commun 8, 2204 (2017). https://doi.org/10.1038/s41467-017-02337-y

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First Josephson Junction (JJ) reservoir (1)

Josephson transmission line (superconducting oscillators) >100GHz

Inputs to selected JJs only

Wave-like reservoir response

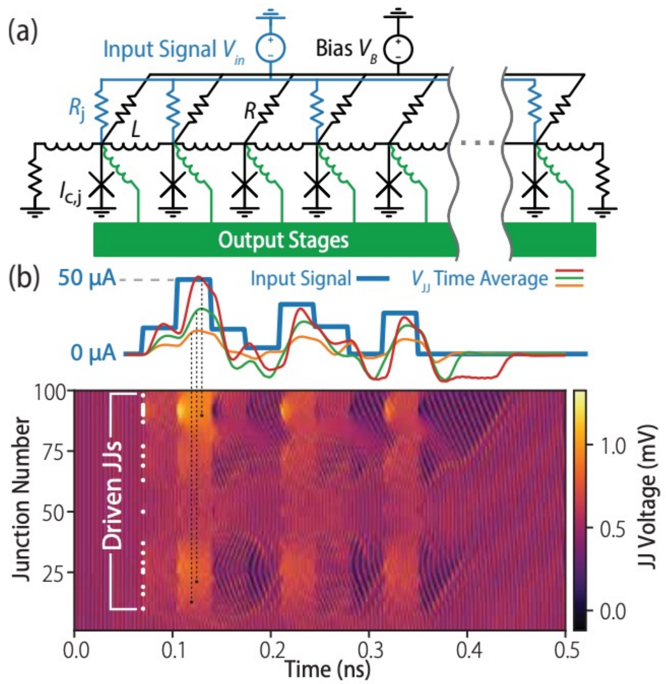


FIG. 1. (a) The superconducting reservoir formed by a Josephson transmission line with the input signal distributed to some fraction f of the junctions across individual resistances R_j. The junctions are connected to decimating output stages. (b) The reservoir response to the displayed 35-ps sample-and-hold input waveform, showing the voltages across each JJ in a chain of 100. Only twenty percent of JJs (white dots) are driven with the input signal shown above. The time-averaged voltages $V_{jj}(t)$ are shown, for a few indicated junctions, superimposed on the input signal.

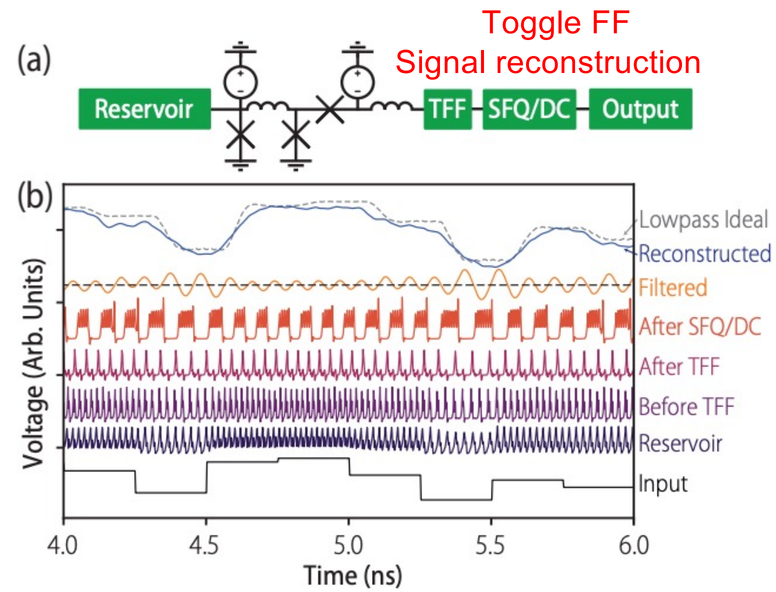
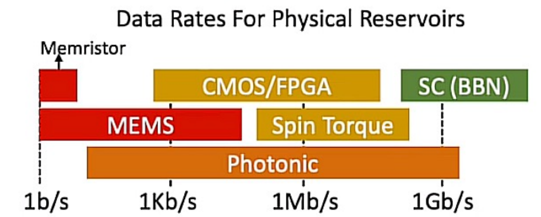


FIG. 2. (a) Detailed view of the output stage, which consists of a buffered output junction, a toggle flip-flop, and an SFQ-to-DC converter. (b) The signal as it evolves, from bottom to top, while passing through the output stage in response to the reservoir stimulus shown at the bottom. The threshold for signal reconstruction is shown as the dotted line at the mean of the "filtered" response and the resulting reconstruction is compared to the "ideal" time-averaged reservoir response.



https://www.youtube.com/watch?v=1bWjyQ1326g

The outputs are the oscillation rates of some subset of the junctions in the reservoir. This is more in line with the liquid state machine approach.

In **simulations** we treat two cases *with similar performance*:

1. **Full rate reservoir**, assuming on-board readout logic. Sample and hold times as short as 15 ps.
2. **Slower reservoir** with additional capacitance and realistic microwave chain for room temperature control and readout. Sample and hold times 250 ps.

Task: speech recognition. ~70% accuracy. 5 JJs.

Rowlands et al., Reservoir Computing with Superconducting Electronics, 2021, https://arxiv.org/abs/2103.02522

Optoelectronic RC (general idea)

Photonic reservoirs are all about high speed. The challenge is to make them small.

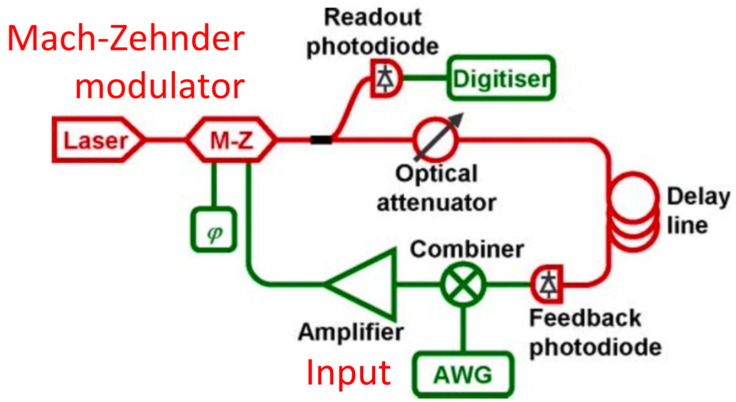


Figure 1 | Schematic of the experimental set-up. The red and green parts represent respectively the optical and electronic components. The optical part of the setup is fiber based, and operates around 1550 nm (standard telecommunication wavelength). "M-Z": Lithium Niobate Mach-Zehnder modulator. "φ": DC voltage determining the operating point of the M-Z modulator. "Combiner" : electronic coupler adding the feedback and input signals. "AWG": arbitrary waveform generator. A computer generates the input signal for a task and feeds it into the system using the arbitrary waveform generator. The response of the system is recorded by a digitiser and retrieved by the computer which optimizes the read-out function in a post processing stage. The feedback gain α is adjusted by changing the average intensity inside the loop with the optical attenuator. The input gain β is adjusted by changing the output voltage of the function generator by a multiplicative factor. The bias φ is adjusted by using a DC voltage to change the operating point of the M-Z modulator. The operation of the system is fully automated and controlled by a computer using MATLAB scripts.

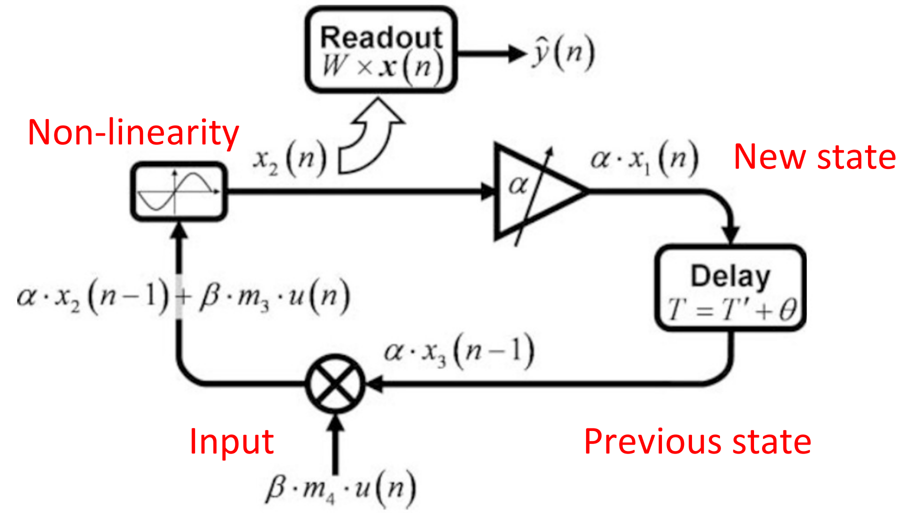


Figure 2 | Schematic diagram of the information flow in the experiment depicted in Fig. 1. On the plot we have represented four reservoir nodes at different stages of processing, labeled according to equation 5 with $k = 1$. Starting from the bottom, and going clockwise, a input value $u(n)$ gets multiplied by an input gain β and a mask value m_i, then mixed with the previous node state $\alpha x_{i-k}(n-1)$. The result goes through the sine function to give the new state of the reservoir $x_i(n)$, which then gets amplified by a factor α and, after the delay, will get mixed with a new input $u(n+1)$. All the network states $x_i(n)$ are also collected by the readout unit, multiplied by their respective weights W_i and added together to give the desired output $\hat{y}(n)$.

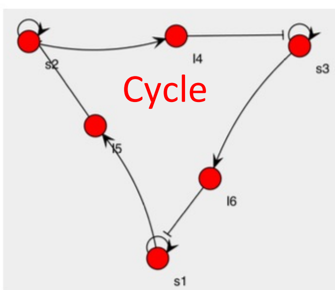
Paquot et al., 2012, http://www.doi.org/10.1038/srep00287

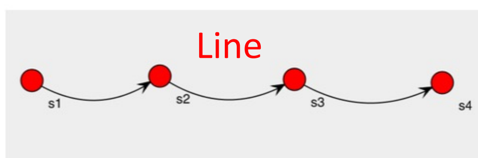
DNA oscillators as reservoirs

$$\frac{d[P_1]}{dt} = h\beta[S_1]([G_1]-[P_3]) - \frac{e}{V}[P_1], \quad \frac{d[S_1]}{dt} = \frac{S_1^m}{V} - h\beta[S_1]([G_1]-[P_3]) - \frac{e}{V}[S_1],$$
$$\frac{d[P_2]}{dt} = h\beta[S_2]([G_2]-[P_1]) - \frac{e}{V}[P_2], \quad \frac{d[S_2]}{dt} = \frac{S_2^m}{V} - h\beta[S_2]([G_2]-[P_1]) - \frac{e}{V}[S_2],$$
$$\frac{d[P_3]}{dt} = h\beta[S_3]([G_3]-[P_2]) - \frac{e}{V}[P_3], \quad \frac{d[S_3]}{dt} = \frac{S_3^m}{V} - h\beta[S_3]([G_3]-[P_2]) - \frac{e}{V}[S_3].$$

Goudarzi, Lakin, and Stefanovic, DNA reservoir computing: a novel molecular computing approach, 2013, https://doi.org/10.1007/978-3-319-01928-4_6

- Yahiro et al., Alife, 2018, https://doi.org/10.1162/isal_a_00013
- Used same oscillator as Goudarzi et al.
- Simulation only. Similar performance as Goudarzi.





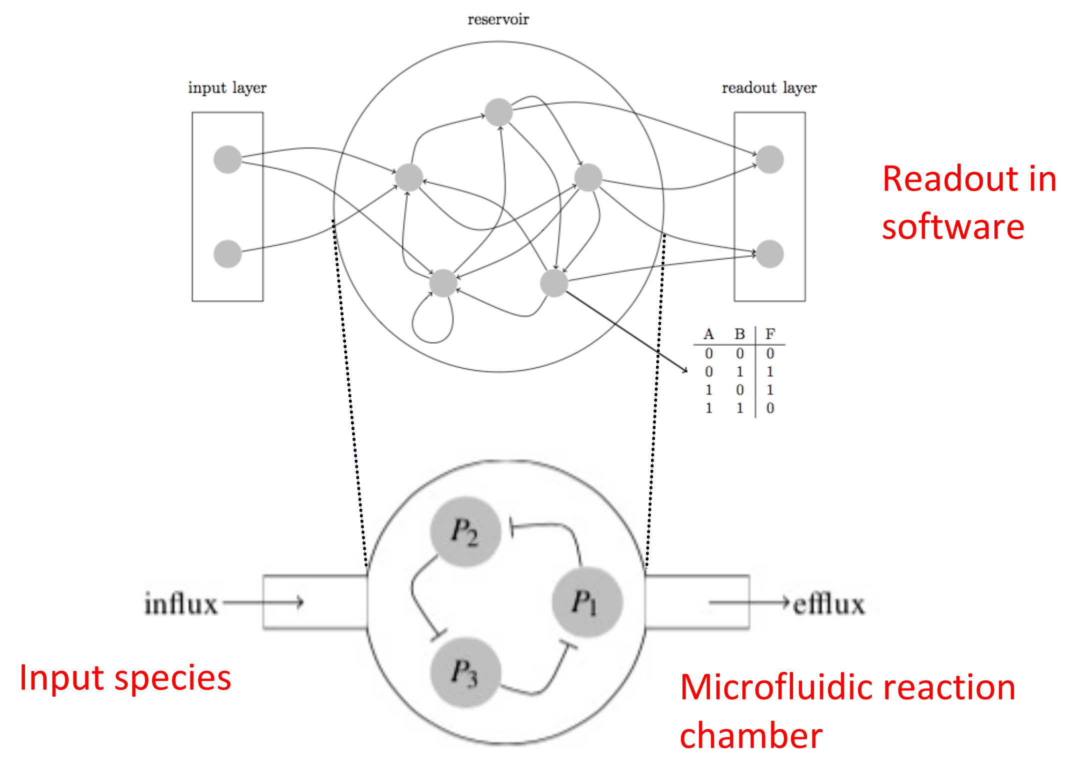
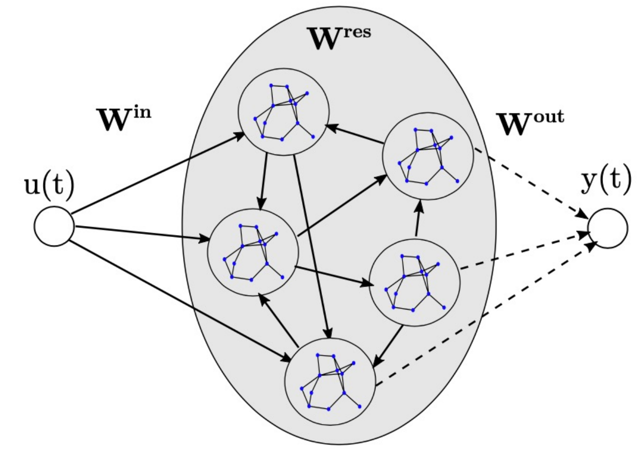
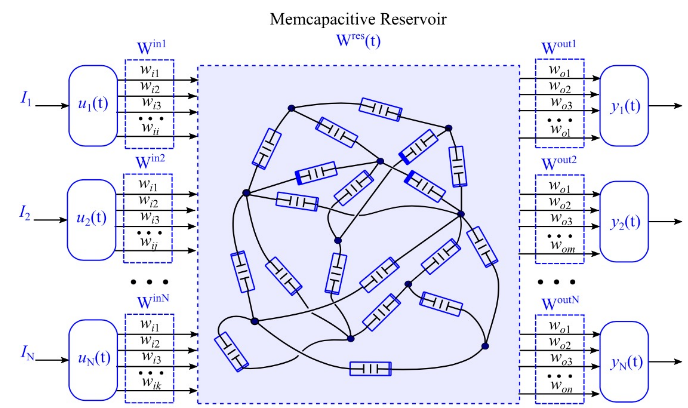


Fig. 4: Three products form an inhibitory cycle that leads to oscillatory behavior in the reservoir. Each product P_i inhibits the production of P_{i+1} by the corresponding deoxyribozyme (cf. Equation 4).

Open problems, challenges, and opportunities

- Scale up PRC systems to **massive** numbers of neurons
- Train **modular/hierarchical** reservoirs efficiently.
- **Multitasking** reservoirs.
- **Lifelong/incremental** learning with growing reservoirs (developmental systems).
- Implement readout/learning directly *in materio*.
- Find **sweet spots** for high-speed, low-power, high performance, and high robustness.
- Reservoirs that adapt, are tuned, or reconfigured (*in-situ*).
- Material reservoir (deep) **co-evolution**.





Quo vadis?

- **Material computing** has attracted increasing attention: it allows for solving computational problems for which traditional, general-purpose architectures are not efficient.

- PRC has great potential for physically “embedding” computing in a substrate by using the substrate itself to perform the computation.

- A broad(er) success of PRC will be contingent on finding solutions to the open questions and challenges: identify physical substrates, scaling up hardware, scaling up programmability → **(deep) co-design**.

To probe further

Neural Networks 115 (2019) 100–123

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Review

Recent advances in physical reservoir computing: A review

Gouhei Tanaka [a,b,*], Toshiyuki Yamane [c], Jean Benoit Héroux [c], Ryosho Nakane [a,b], Naoki Kanazawa [c], Seiji Takeda [c], Hidetoshi Numata [c], Daiju Nakano [c], Akira Hirose [a,b]

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ABSTRACT

Reservoir computing is a computational framework suited for temporal/sequential data processing. It is derived from several recurrent neural network models, including echo state networks and liquid state machines. A reservoir computing system consists of a reservoir for mapping inputs into a high-dimensional space and a readout for pattern analysis from the high-dimensional states in the reservoir. The reservoir is fixed and only the readout is trained with a simple method such as linear regression and classification. Thus, the major advantage of reservoir computing compared to other recurrent neural networks is fast learning, resulting in low training cost. Another advantage is that the reservoir without adaptive updating is amenable to hardware implementation using a variety of physical systems, substrates, and devices. In fact, such physical reservoir computing has attracted increasing attention in diverse fields of research. The purpose of this review is to provide an overview of recent advances in physical reservoir computing by classifying them according to the type of the reservoir. We discuss the current issues and perspectives related to physical reservoir computing, in order to further expand its practical applications and develop next-generation machine learning systems.

https://doi.org/10.1016/j.neunet.2019.03.005

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5 August 2022

TOPICAL REVIEW

Hands-on reservoir computing: a tutorial for practical implementation

Matteo Cucchi[1,2,6], Steven Abreu[3,4,6], Giuseppe Ciccone[1], Daniel Brunner[5] and Hans Kleemann[1,*]

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2 Laboratory for Soft Bioelectronic Interfaces, Ecole Polytechnique Fédérale de Lausanne (EPFL), Geneva, Switzerland
3 Bernoulli Institute for Mathematics, Computer Science and Artificial Intelligence & Cognitive Systems and Materials Center (CogniGron), University of Groningen, The Netherlands
4 Institute of Neuroinformatics, University of Zurich and ETH Zurich, Switzerland
5 FEMTO-ST/Optics Dept., UMR CNRS 6174, Univ. Bourgogne Franche-Comté, 15B avenue des Montboucons, 25030 Besançon Cedex, France
* Author to whom any correspondence should be addressed.
6 These authors contributed equally to this work.

https://doi.org/10.1088/2634-4386/ac7db7

Deep co-design: our best path forward?

Given a proposed unconventional computing substrate, we can ask:
- Does it actually compute?
- If so, how well does it compute?
- Can it be made to compute better?

Given a proposed unconventional computational model, we can ask:
- How powerful is the model?
- Can it be implemented in a substrate?
- How faithfully or efficiently can it be implemented?

Given complete freedom in the choice of model and substrate, we can ask:
- Can we co-design a model and substrate to work well together?

Stepney, S., Co-designing the computational model and the computing substrate. In Proceedings of the 18th International Unconventional Computation and Natural Computation Conference, pp 5-14, 2019.

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Deep co-design across the entire stack?

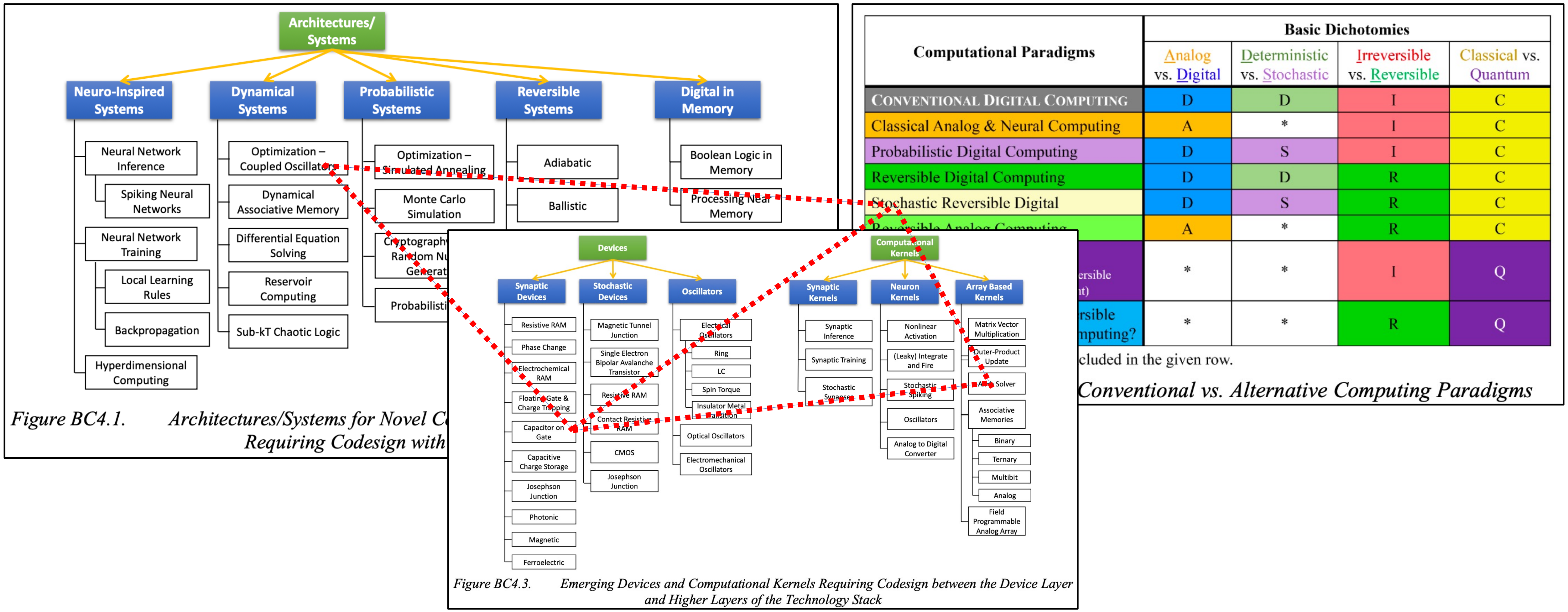

Figure BC4.3. Emerging Devices and Computational Kernels Requiring Codesign between the Device Layer and Higher Layers of the Technology Stack

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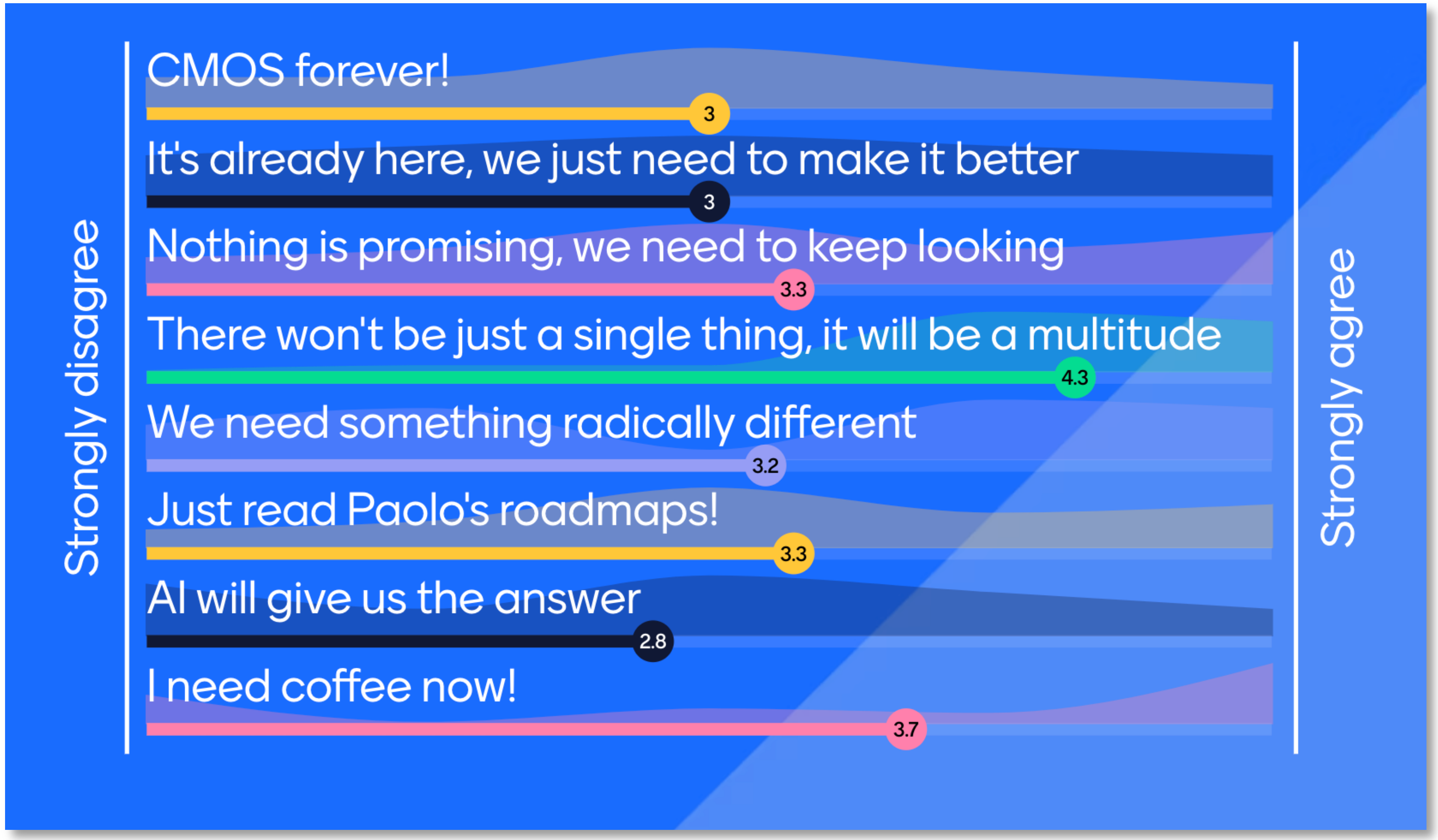
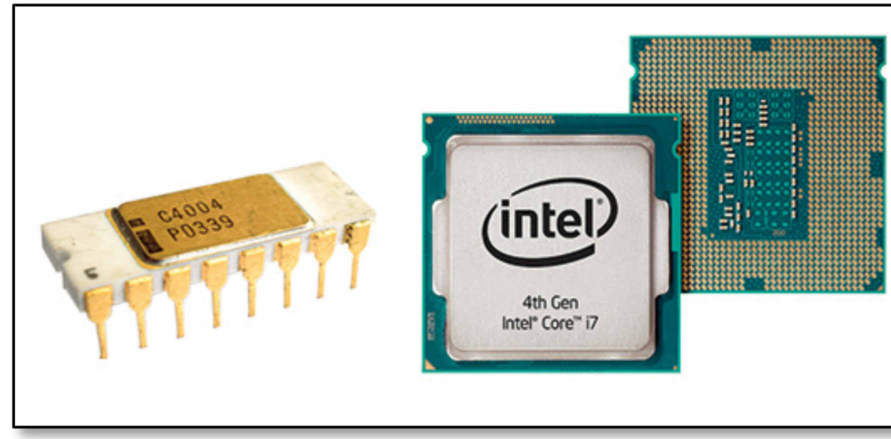
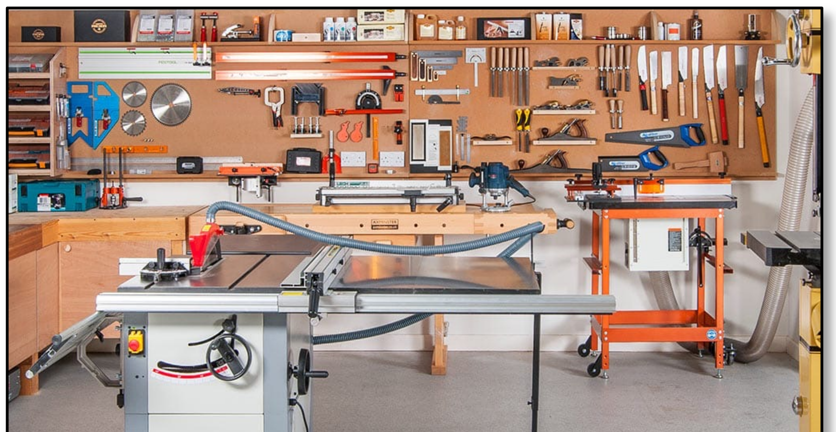
Where should we look for the next big thing?



ICRC 2023 poll

Look for specialists, not generalists





Action items for the community

- Look for specialists
- Formal frameworks
- (Unified?) evaluation metrics
- Roadmap(s)
- Timelines to solutions

Scott Holmes

The one and only take-home message?

Compute is cheap, moving bits is expensive.

What may be next?

- **Applications — AI outside of the datacenter:**
  - Must be cheap, fast, and efficient. cheaper to run and easier to deploy.
  - E.g., smartphones that run advanced language models without draining the battery, or sensors that learn locally without sending every bit of data to a datacenter.
- **Architecture — Memory and interconnect:**
  - The next decade of AI hardware is mostly a memory and interconnect story, not a compute story. Whoever solves bandwidth, locality, and energy-per-bit-moved wins. The exotic devices (memristors, photonics, Mott materials) matter most where they collapse the memory–compute distance, not where they do clever math.
- **EDA — Mostly to fully autonomous design and verification across the entire stack**
  - Agentic design, agentic verification, agentic everything…

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brainchip AKD1500 Edge AI Co-Processor Product Brief

The AKD1500 is an Edge AI acceleration co-processor built on the Akida™ Neuromorphic processing engine. It accelerates most available neural network models using a highly energy-efficient, event-based, and purely digital, neuromorphic processing architecture. It has built-in capabilities to learn on-device, enabling secure application personalization, without the need for a Cloud connection or retraining. BrainChip's unique MetaTF™ software flow enables developers to compile and optimize their chosen models for the AKD1500. The Intelligent Runtime software manages network processing to fully utilize available resources and automatically partitions execution into multiple passes if resources are insufficient. This enables today's implementations to accelerate tomorrow's models. The AKD1500 can be paired with any host CPU or MCU via PCIe or low-power SPI interfaces to create very compact, low-power, portable and intelligent devices for Internet of things (IoT), Industrial, Automotive, Healthcare, Consumer, Smart Home and Smart City applications. Furthermore, it has an ideal form-factor for MCU and MPU modules.

Key Benefits

- **Designed for Low-Power AI NN Acceleration**
  - Unsurpassed images/second/watt
  - Maximum 800 Effective GOPS at <1mW/GOPS
- **On-Chip Learning**
  - Personalized Edge AI systems
- **Industry Standard Development Environment**
  - TensorFlow/Keras and Pytorch APIs
- **Cost Effective**
  - 22 nm FD-SOI CMOS digital logic process
  - 7x7 mm MFCTFBGA169 package, 0.5 mm pitch

Example Applications Areas

- **Personalized Learning Edge AI systems**
- **Edge AI Vision Systems**
  - ADAS/AV
  - Vision Guided Robotics
  - Drones
  - Video Surveillance
- **Industrial Internet of Things**
  - Environmental monitoring/control
  - Predictive maintenance
- **Smart Home**
  - Appliances
  - Speakers
  - Voice Control

Specifications

- Akida Neuron Fabric
- On-Chip Conversion Complex
- 1 MB On-Chip Local memory
- PCIe Gen2 Endpoint Interface
- SPI S/D/Q/O Peripheral Interface
- SPI D/Q/O Memory Expansion Interface
- Clock frequency range: 5 – 400 MHz
- Power dissipation: 250 mW typical at 400 MHz

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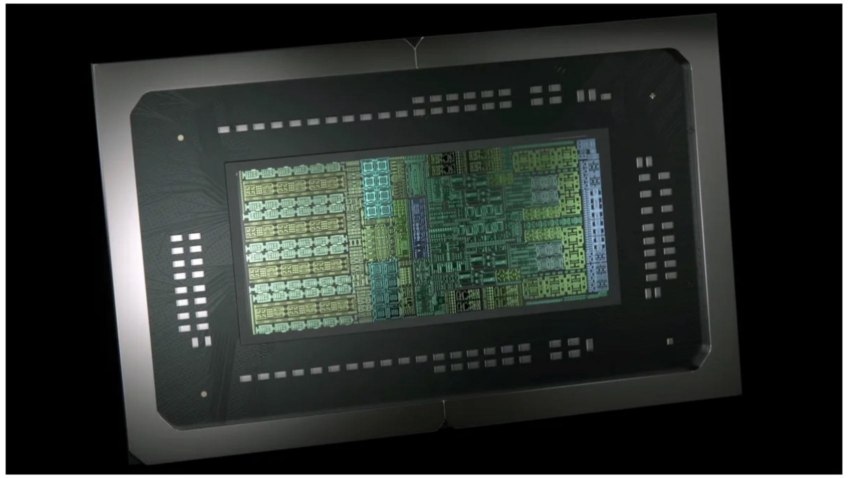
Nvidia's long-awaited N1/N1X SoC specs leak ahead of Computex launch — N1 to feature up to 20 Arm-based cores, standard N1 equipped with 12- and 10-core configs

News By Hassam Nasir published yesterday

The top-end N1X SKU is essentially a rebranded GB10.



(Image credit: Nvidia)

teuscher Lab teuscher-lab.com

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Near-term (1–3 yrs): formats, sparsity, and the memory wall

Already shipping or close — wins come from data movement and number formats, not exotic devices

Low-precision + sparsity

- FP4 / MXFP4, block-scaled formats
- 2:4 / N:M structured sparsity
- Mainline in Blackwell, MI300, Groq
- Easy perf/W wins for 2 generations

Memory bandwidth is binding

- HBM4, 3D-stacked SRAM (X3D-style)
- On-package optical I/O emerging
- Training is a memory-system problem
- FLOPs are no longer the bottleneck

Inference ASICs take share

- Groq, Etched, SambaNova, MTIA, Trainium2
- Transformer-tuned silicon
- Beat GPUs on tokens / $ / W
- GPUs stay king for training

Medium-term (3–7 yrs): photons, in-memory compute, 3D packaging

High-confidence shifts — collapsing the distance between memory and compute

Co-packaged optics

- Electrical SerDes hit bandwidth wall
- Silicon photonics chip-to-chip / rack
- Lightmatter, Ayar Labs, Celestial
- Optical interconnect ≫ optical compute

Analog in-memory compute

- Crossbar arrays for edge inference
- PCM, RRAM, MRAM, nickelates
- 4-bit accuracy is enough for inference
- IBM analog AI, EnCharge, Mythic

Wafer-scale + 3D stacking

- Cerebras proved wafer-scale works
- TSMC CoWoS / SoIC roadmaps
- Logic-on-logic, logic-on-memory
- Default packaging story by ~2030

Long-term (7+ yrs): bets, not forecasts

Cross-cutting theme: compute is cheap — moving bits is expensive

Neuromorphic at the edge

- Loihi-class, memristive spiking arrays
- Sensor fusion, robotics, BMI
- Event-driven sparsity wins
- Parallel market, not LLM replacement

Mott / phase-change substrates

- One material → R, C, neuron, synapse
- Strongly-correlated oxides (e.g. NNO)
- Collapses memory-compute distance
- Path from 'interesting' to 'deployed'

Cryo-CMOS / superconducting

- Gigawatt-scale data center power untenable
- Less sexy than quantum, more likely
- Quantum ML: skeptical this decade
- Quantum-inspired classical more useful

What is likely hyped?

- **Photonic matrix multiply** as a near-term GPU replacement. Modulator energy and ADC overhead still dominate.
- **General-purpose neuromorphic for LLMs**. Spiking doesn't map cleanly onto attention.
- **Pure analog training**. Inference, yes. Training needs precision that analog can't sustain through millions of updates without drift correction that eats the energy savings.

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How do you prepare and survive in this?

Bet on durable skills, not on whichever device or framework is hot this year

Go deep on the stack

- Pick one layer cold: device, circuit, arch, or system
- Read fluently one layer above and below
- Interesting work lives at the interfaces
- Cross-layer engineers are rare and paid well
- Master co-design

Master the fundamentals

- Linear algebra, numerics, probability
- Caches, memory hierarchy, pipelines, NoCs
- Roofline analysis
- Power, thermals, packaging
- Solid-state physics for emerging devices

Build, don't just study

- Tape out on Tiny Tapeout / OpenROAD
- Train + quantize a model on real hardware
- Contribute to open silicon (Chipyard, MLIR)
- A GitHub history beats a transcript

Pick durable bets, hedge the rest

- Bandwidth, energy/bit, locality always matter
- Architectures turn over; the math doesn't
- Track 2–3 emerging directions seriously
- Be willing to be wrong, adapt quickly